

AEC / SDC · SEMESTER III · PROJECT-BASED LEARNING · ONE CREDIT

Complete Community-Project Notes with the Tool It Documents

COMMUNITY PROJECT

Topic 1 — Computer-Based Learning Tool for Mentally
Challenged Students · Mangaluru, Dakshina Kannada · Fifteen
Week-Led Chapters, One per TLP Row · Every Number
Captured from Programs Run Twice, Byte-Identical · Offline
Browser Tool, Fingerprinted Content Pack, Pilot Data, DPR
Ledger · Viva Drill Included

COURSE CODE

1BCP308

Credits	Teaching Hours (L : T : P)	Total Marks	CIE	SEE	Exam Hours	Examination Type
1	0:0:0	100	50	50	03 Hours	Practical

by

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Course at a Glance

Course	Community Project / Societal Project (Project-Based Learning)
Course Code	1BCP308 (AEC/SDC, Semester III, 1 credit)
Pedagogy ledger	0:0:0:30:30 — 0 lecture, 0 tutorial, 0 lab contact; 30 h term work + 30 h self-learning. The contact hours are in the weeks, and the weeks are the table of contents of this book.
The one topic developed here	Computer-Based Learning Tool for Mentally Challenged Students — district Dakshina Kannada, Mangaluru, Karnataka. Tagged PO5 & PO 6 and SDG 4.
Desired outcome	Design and develop a web/browser-based learning tool as per the requirements of the special school.
Assessment	CIE 50 (committee of three, six rubric rows 8/10/8/10/6/8) · SEE 100 scaled to 50 (six rows 16/20/16/20/12/16) · gates 20/50, 35/100, total 40
Environment of record	Python 3.13.14 · numpy 2.3.5 · pandas 2.2.3 on Debian GNU/Linux 13 (trixie) — RESULT: toolchain captured
What this book is not	Not a tutorial on web development. It is the audit trail of a semester: fifteen programs, one static tool, three CSVs, and a book whose every number is a quote.

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Preface

This volume follows one rule all semester: nothing is hand-typed. The syllabus was captured once into a data module; every rubric weight, gate, week row and requirement in these pages is either printed from that module or quoted from a program transcript. The fifteen chapters are the fifteen weeks of the Teaching-Learning Process for the one project topic this edition develops — a computer-based learning tool for mentally challenged students, built for and with three special schools of Dakshina Kannada.

Each chapter contains the week's program, the terminal session it produced, and a set of evidence chips: lines quoted verbatim from the capture. Between the two capture passes, every out/ file was byte-compared; the two runs matched. When a chapter makes a claim — a slope, a fingerprint, a contrast ratio, a mark ceiling — you can find the exact line it came from, and you can recompute it by running the program yourself.

It is written for the team that must defend this project in a viva and for the committee that must grade it. Chapters 1 to 8 build the case; 9 to 15 build and prove the tool. The appendices hold the rubrics as printed, the viva bank, the reproduction pipeline, and a debug log of what the discipline caught — because a methodology that never bites was never enforced.

Machine-verified anchors of this edition: RESULT: capture: 17 programs ok — engine fingerprint — hash-verified.

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Prescribed Course Content — Verbatim

Everything in this block is transcribed from the captured course page (docs/syllabus-capture.md) by way of `syllabus_data.py`; it is printed, not paraphrased.

Course objective

The Community Project / Societal Project is a practical, project-based learning course that builds students' awareness on Educational Social Responsibility (ESR). It focuses on real issues such as the adaptation of technology for rural and urban development, the application of tools for enhanced governance in civic societies, and sustainable technological solutions for better living. Students may work individually or in groups of up to four, including interdisciplinary topics, and assessment is based on the prescribed rubrics.

How the project shall be conducted

Societal projects should be academic, socially responsible, technically focused, and politically neutral. Students and faculty must maintain professionalism, ethical conduct, and respect while working with communities and stakeholders and avoid any political, ideological, commercial, or disruptive activities. Projects should also consider practical constraints such as cost, resources, safety, social impact, and stakeholder cooperation.

Method of community engagement

Interaction with local rural or urban stakeholders such as local leaders, corporation/municipality/Panchayath officers, NGOs, etc.; review of issues highlighted in print and electronic media and social networks.

After selecting a broad topic, collect data and requirements, conduct a feasibility survey, estimate timelines and resources, approximate expenditure (if applicable), and define the problem statement with objectives clearly.

The project topic developed in this volume (row 1 of the course list)

Computer-Based Learning Tool for Mentally Challenged Students — Mangaluru, Dakshina Karnataka, Karnataka. Outcome: Design and develop a web/browser-based learning tool as per the requirements of the special school. Program Outcomes PO5, 6 · SDG 4.

Teaching-Learning Process (15 weeks) — the syllabus's own table

Wk	Activity	Faculty role	Student activity	Output / evidence	Chapter
1	Course introduction, objectives, ethics, conduct, and assessment briefing	Explain course scope, rules, rubrics, and expected deliverables	Understand course requirements and form teams	Team list, course plan	1
2	Community problem identification and topic selection	Guide	Topic Framing and Feasibility — Identify local issues through discussion and observation	Shortlisted project topic	2
3	Stakeholder interaction and preliminary survey	Demonstrate survey methods and stakeholder engagement	Collect initial data from community/users	Survey notes, interview record	3
4	Problem statement and objective formulation	Review and refine	Problem statement, scope, and objectives	Approved problem statement	4
5	Literature/case study review and baseline analysis	Suggest relevant references and examples	Study existing solutions and compare approaches	Baseline analysis summary	5
6	Project planning and role assignment	Help teams organize tasks, timeline, and responsibilities	Plan activities, divide roles, and prepare schedule	Work plan / Gantt chart	6
7	Feasibility analysis and solution brainstorming	Check technical feasibility and practicality	Generate possible engineering solutions	Solution alternatives sheet	7
8					8

Wk	Activity	Faculty role	Student activity	Output / evidence	Chapter
	Selection of final solution and design approach	Review final concept and design logic	Finalize concept, methods, and expected outcome	Concept note / design outline	
9	Data processing / calculations / sizing / mapping	Support Computations and technical review	Analyze data, perform calculations, prepare maps/charts	Technical analysis sheet	9
10	Prototype / model / audit / DPR / app development	Monitor progress and correct technical issues	Develop the selected output	Working model/ draft DPR / map/ app	10
11	Testing, refinement, and implementation planning	Evaluate technical correctness and improvements	Test prototype or validate proposal with users	Test results / revised output	11
12	Field implementation / demonstration / awareness activity	Facilitate community interaction and documentation	Conduct implementation or demonstration	Photos, attendance, implementation record	12
13	Documentation of outcomes, feedback, and impact	Review report structure and completeness	Compile findings, stakeholder feedback, and observations	Draft report	13
14	Presentation preparation and viva practice	Conduct mock presentation and feedback	Prepare slides/poster and rehearse	Presentation deck	14
15	Final presentation, submission, and reflection	Evaluate performance using rubrics	Present project, submit report, reflect on learning	Final report, presentation, viva	15

Assessment rules, verbatim

CIE $\geq 20/50$ (minimum) to become eligible to appear for the SEE.

SEE $\geq 35/100$ in the SEE itself to be eligible for pass consideration.

CIE (out of 50) + SEE (scaled to 50) ≥ 40 for a pass grade; otherwise grade F (Fail).

CIE pass requirement: A student shall secure a minimum of 20 marks out of 50 in CIE to become eligible to appear for SEE. For a pass in SEE, a student shall secure minimum 35 marks out of 100. For declaration of a pass grade, the sum of CIE marks (out of 50) and scaled SEE marks (out of 50) shall be ≥ 40 ; otherwise, the student is awarded grade F (Fail).

CIE will be conducted by a departmental committee (HoD/nominee, guide and one senior faculty) using the following rubrics totalling 50 marks. The SEE shall be conducted batch-wise by a committee of three faculty: project guide, one faculty from the same department and one from a different department. Maximum examination marks: 100 (entered as 100 in the VTU portal; the ITI-SMU module automatically scales 100 $\rightarrow$ 50 for result processing).

CIE rubric — six criteria, 50 marks

Criterion (weight 50)	Excellent	Good	Fair	Needs Improvement
Problem identification & relevance to society/ environment (8)	8 · Problem clearly identified with strong evidence of relevance and well-defined scope.	6 · Relevant problem with clear scope and adequate evidence.	4 · Problem defined but justification is weak.	2 · Problem unclear or not relevant.
Formation of problem statement; engineering approach, innovation & feasibility (10)	10 · Precise statement; sound, innovative, feasible approach.	8 · Adequate statement; feasible with minor gaps.	5 · Partial statement; marginal feasibility.	2 · Missing or infeasible.
Interaction with stakeholders; data collection, analysis & interpretation (8)	8 · Thorough engagement; comprehensive data and well-interpreted analysis.	6 · Good engagement; adequate data and analysis.	4 · Limited engagement; incomplete data and superficial analysis.	2 · Little or no stakeholder contact; no data/ analysis.
Project documentation; clarity, organization & problem-solving demonstrated (10)	10 · Complete, well-organized, analytical, and strong problem-solving.	8 · Complete and clear with some analytical content.	5 · Basic with omissions; problem-solving limited.	2 · Missing/poorly organized; no clear problem-solving.
Communication & presentation skills (6)	6 · Confident, structured, and	5 · Clear and satisfactory;	3 · Acceptable but lacks clarity or	2 · Poor or absent; unable to respond.

Criterion (weight 50)	Excellent	Good	Fair	Needs Improvement
	answers queries fully; visuals effective.	responds to most queries.	confidence; weak responses.	
Contribution & benefit to society (impact, sustainability) (8)	8 · Clear, demonstrable societal benefit and realistic sustainability path.	6 · Benefit plausible; sustainability considered with minor gaps.	4 · Limited or unclear benefit; sustainability weak.	2 · No evident benefit or sustainability.

SEE rubric — six criteria, 100 marks (scaled to 50)

Criterion (weight 100)	Excellent	Good	Fair	Needs Improvement
Problem identification & relevance to society/environment (16)	16 · Clearly articulated with strong evidence and context.	12 · Clear with reasonable evidence.	8 · Problem stated but evidence/context weak.	4 · Poorly defined or irrelevant.
Formation of problem statement; engineering approach, innovation & feasibility (20)	20 · Precise; demonstrates engineering depth, novelty and feasibility.	15 · Technically sound and feasible with minor limitations.	10 · Basic; limited technical depth or feasibility concerns.	5 · No viable engineering approach or infeasible plan.
Interaction with stakeholders; data collection, analysis & interpretation (16)	16 · Excellent engagement; rigorous data collection and insightful analysis.	12 · Good engagement and acceptable analysis.	8 · Superficial data collection and limited analysis.	4 · Inadequate or no data; poor analysis.
Project documentation; clarity, technical rigor & problem-solving demonstrated (20)	20 · Comprehensive, technically rigorous, and systematic.	15 · Complete and technically correct with minor gaps.	10 · Lacks technical depth or has notable omissions.	5 · Missing key elements or technically incorrect.
Communication, presentation & viva-voce responses (12)	12 · Presents expertly, defends work persuasively and answers all queries correctly.	9 · Good presentation and satisfactory answers to most queries.	6 · Adequate but fails to answer some technical questions.	3 · Poor presentation and unable to defend work.
Contribution & benefit to society; impact assessment & sustainability (16)	16 · Clear, evidence-based impact assessment with scaling recommendations.	12 · Reasonable impact discussion and sustainability considerations.	8 · Limited evaluation of impact or sustainability.	4 · No coherent assessment of impact or sustainability.

FRONT MATTER

Compliance Maps & Assessment Arithmetic

Chapter → program → evidence → course outcome → TLP output

Ch	Chapter	Capture	COs	Required output (syllabus)
Ch 1	The Briefing Ledger: What the Course Actually Pays For	out/cp00_course_brief.txt	CO1, CO2	Team list, course plan
Ch 2	Screening the Topic Against the Ground It Must Land On	out/cp01_topic_screen.txt	CO1, CO5	Shortlisted project topic
Ch 3	The Baseline Survey: 48 Respondents, Three Schools	out/cp02_stakeholder_survey.txt	CO1, CO2, CO4	Survey notes, interview record
Ch 4	Freezing the Problem Statement on Evidence Anchors	out/cp03_problem_statement.txt	CO1, CO2	Approved problem statement
Ch 5	The Baseline: What Already Exists, and What It Misses	out/cp04_literature_baseline.txt	CO1, CO5	Baseline analysis summary
Ch 6	The Week Map: Fifteen Slots, One Float, Loads Accounted	out/cp05_plan_gantt.txt	CO2	Work plan / Gantt chart
Ch 7	Feasibility: What the School Already Owns	out/cp06_feasibility_cost.txt	CO5	Solution alternatives sheet
Ch 8	Design Week: The Engine Surface and the Page Graph	out/cp07_concept_design.txt	CO2, CO6	Concept note / design outline
Ch 9	The Content Pack: Sixty-seven Items on a Difficulty Ladder	out/cp08_content_pack.txt	CO2, CO6	Technical analysis sheet
Ch 10	The Build: Six Files, Zero Network Calls	out/cp09_build_tool.txt	CO6	Working model/draft DPR / map/app
Ch 11	Testing: a Truth Table for the Engine, a Receipt for the Classroom	out/cp10_testing.txt	CO2, CO4	Test results / revised output
Ch 12	The Pilot: Twelve Weeks of Usage, Read Without Mercy	out/cp11_pilot_analysis.txt	CO4	Photos, attendance, implementation record
Ch 13	Impact and the DPR: Numbers That Cross-Check Across Files	out/cp12_impact_dpr.txt	CO3, CO4, CO6	Draft report
Ch 14	The Evidence Check: Recompute Everything, Compare Bytes	out/cp13_evidence_check.txt	CO2, CO4	Presentation deck
Ch 15	The Viva Ledger: Every Answer Regenerated From Disk	out/cp14_viva_ledger.txt	CO4, CO6	Final report, presentation, viva

CIE rubric rows → the chapters that generate their evidence

CIE criterion	Where proven
Problem identification & relevance to society/environment (8)	Ch 1, Ch 2
Formation of problem statement; engineering approach, innovation & feasibility (10)	Ch 4, Ch 6, Ch 7, Ch 8
Interaction with stakeholders; data collection, analysis & interpretation (8)	Ch 3, Ch 4
Project documentation; clarity, organization & problem-solving demonstrated (10)	Ch 13
Communication & presentation skills (6)	Ch 15
Contribution & benefit to society (impact, sustainability) (8)	Ch 12, Ch 13

SEE rubric rows → the chapters a committee will cross-examine

SEE criterion	Where defended
Problem identification & relevance to society/environment (16)	Ch 1, Ch 2, Ch 4
Formation of problem statement; engineering approach, innovation & feasibility (20)	Ch 8, Ch 9, Ch 10
Interaction with stakeholders; data collection, analysis & interpretation (16)	Ch 3, Ch 12
Project documentation; clarity, technical rigor & problem-solving demonstrated (20)	Ch 13, Ch 14

SEE criterion	Where defended
Communication, presentation & viva-voce responses (12)	Ch 15
Contribution & benefit to society; impact assessment & sustainability (16)	Ch 12, Ch 13

Course outcomes

CO	Level	Statement
CO1	L2 – Understand	Identify and analyze societal problems through field interaction and technical assessment.
CO2	L3 – Apply	Develop skills in project planning, field survey, data collection, implementation, documentation, and reporting.
CO3	L3 – Apply	Apply principles of social responsibility in engineering practice through community-oriented interventions
CO4	L4 – Analyze	Exhibit multidisciplinary learning, problem-solving ability, and communication skills based on collected data (requirements), stakeholder feedback, and baseline conditions.
CO5	L5 – Evaluate	Choose feasible societal problem statements and solutions considering practical constraints and sustainability aspects.
CO6	L6 – Create	Create a product by collaborating with stakeholders and within the team to deliver a product/project to the intended community/organization.

Program outcomes and the sustainable development goal on this topic's line

PO	Statement
PO5	Modern tool usage: create, select, and apply appropriate techniques, resources, and modern IT and engineering tools to complex engineering activities, with an understanding of the limitations.
PO6	The engineer and society: assess societal, health, safety, legal, and cultural issues and the relevance of professional engineering work.
SDG	Quality Education (SDG 4) — ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

The arithmetic every submission must survive

CP-00 RESULT: rubric arithmetic verified 50/100, gates 20/50, 35/100, 40 total – worst-mix SEE raw 40 – restated from source -- PASS

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P A R T



Method — Chapters 1 to 8

The first eight weeks of the syllabus's Teaching-Learning Process, one chapter each: what the course pays for, why this topic, what the schools said, what the problem statement froze, what already exists, how fifteen weeks were planned, what the money test returned, and how the design was made auditable. Every number in this part is a quote from a capture, and every capture was run twice.

IN THIS PART

- Chapter 1 — The Briefing Ledger: What the Course Actually Pays For
- Chapter 2 — Screening the Topic Against the Ground It Must Land On
- Chapter 3 — The Baseline Survey: 48 Respondents, Three Schools
- Chapter 4 — Freezing the Problem Statement on Evidence Anchors
- Chapter 5 — The Baseline: What Already Exists, and What It Misses
- Chapter 6 — The Week Map: Fifteen Slots, One Float, Loads Accounted
- Chapter 7 — Feasibility: What the School Already Owns
- Chapter 8 — Design Week: The Engine Surface and the Page Graph

Chapter 1 — The Briefing Ledger: What the Course Actually Pays For

WEEK (TLP ROW)	1 — Course introduction, objectives, ethics, conduct, and assessment briefing · students: Understand course requirements and form teams · evidence: Team list, course plan
FILES	src/cp00_course_brief.py, out/cp00_course_brief.txt
COURSE OUTCOMES	CO1, CO2 (see the maps in the front matter)

A credit with zero contact hours is either free or it is read carefully. 1BCP308 is read carefully: the ledger lives in the Teaching-Learning Process rows, and the money lives in two rubric tables. Week 1 therefore produces no slides and no opinion — it produces a program that sums the rubric weights, regex-parses the three pass gates out of the rules text, and proves the arithmetic closes. If the briefing slide and the rubric file ever disagree, this program fails at build time.

1.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

1.2 The Program (Code)

The listing below is the file executed in §1.3 — same bytes, same run.

```
src/cp00_course_brief.py · 44 lines, max width 129 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-00 (week 1): course briefing ledger — rules, rubric arithmetic, gates.
Everything printed here is computed from syllabus_data.py (the captured course page),
so the briefing slide and this transcript can never disagree."""
import os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, ROOT)
import syllabus_data as S

print("== CP-00 Week-1 briefing: what the course pays for, verified from its own text ==")
print()
w_cie = [r[1] for r in S.RUB_CIE]
w_see = [r[1] for r in S.RUB_SEE]
print("CIE rubric weights  :", " + ".join(str(w) for w in w_cie), "=", sum(w_cie), "(must be 50)")
print("SEE rubric weights  :", " + ".join(str(w) for w in w_see), "=", sum(w_see), "(entered /100)")
print("SEE scaling          : 100 -> 50 (ITI-SMU module, automatic)")
assert sum(w_cie) == 50 and sum(w_see) == 100, "rubric weights do not sum"
print()
import re
t = " ".join(S.ASSESS["rules"])
g_cie = int(re.search(r"CIE . (\d+)/(\d+)", t)[1]); g_cie_max = int(re.search(r"CIE . \d+/(\d+)", t)[1])
m_see = re.search(r"SEE . (\d+)/(\d+)", t); g_see = int(m_see[1]); g_see_max = int(m_see[2])
m_pass = re.search(r"scaled to (\d+)\. (\d+)", t); g_scaled_max = int(m_pass[1]); g_pass = int(m_pass[2])
assert g_cie_max == sum(w_cie) and g_see_max == 100, "gate maxima disagree with rubric sums"
need_scaled = max(g_pass - g_cie, 0)
need_raw = max(g_see, -(-need_scaled * g_see_max // g_scaled_max))
print("gates (computed from the rules strings by regex, not remembered):")
print(f" CIE eligibility   : >= {g_cie} of {g_cie_max} = {g_cie/g_cie_max:.0%} of CIE")
print(f" SEE pass floor    : >= {g_see} of {g_see_max} in the SEE itself")
print(f" overall pass      : CIE/{g_cie_max} + SEE scaled/{g_scaled_max} >= {g_pass}, else grade F")
print(f" worst passing mix : CIE {g_cie} + SEE {need_scaled} scaled -> SEE raw {need_raw}/{g_see_max} is the floor at bare CIE")
print()
print("team rule : batches of 3-4, at least one project (or two related activities)")
print("conduct   : academic, socially responsible, technically focused, politically neutral")
print("ledger     : L:T:P:SL:TW =", S.HDR["pedagogy"], "-> 30 h TW + 30 h SL, zero contact hours")
print()
print("course outcomes on the board (level : verb):")
for cid, lvl, _ in S.COS:
    print(f" {cid} ({lvl})")
```

```
print()
print(f"CP-00 RESULT: rubric arithmetic verified {sum(w_cie)}/{sum(w_see)}, gates "
      f"{g_cie}/{g_cie_max}, {g_see}/{g_see_max}, {g_pass} total – worst-mix SEE raw "
      f"{need_raw} – restated from source -- PASS")
```

1.3 Captured Run (Two-Pass Verified)

cp00_course_brief

```
$ python3 src/cp00_course_brief.py
== CP-00 Week-1 briefing: what the course pays for, verified from its own text ==

CIE rubric weights : 8 + 10 + 8 + 10 + 6 + 8 = 50 (must be 50)
SEE rubric weights : 16 + 20 + 16 + 20 + 12 + 16 = 100 (entered /100)
SEE scaling       : 100 -> 50 (ITI-SMU module, automatic)

gates (computed from the rules strings by regex, not remembered):
  CIE eligibility : >= 20 of 50 = 40% of CIE
  SEE pass floor  : >= 35 of 100 in the SEE itself
  overall pass    : CIE(/50) + SEE scaled(/50) >= 40, else grade F
  worst passing mix : CIE 20 + SEE 20 scaled -> SEE raw 40/100 is the floor at bare CIE

team rule : batches of 3-4, at least one project (or two related activities)
conduct   : academic, socially responsible, technically focused, politically neutral
ledger    : L:T:P:SL:TW = 0:0:0:30:30 -> 30 h TW + 30 h SL, zero contact hours

course outcomes on the board (level : verb):
  C01 (L2 – Understand)
  C02 (L3 – Apply)
  C03 (L3 – Apply)
  C04 (L4 – Analyze)
  C05 (L5 – Evaluate)
  C06 (L6 – Create)

CP-00 RESULT: rubric arithmetic verified 50/100, gates 20/50, 35/100, 40 total – worst-mix SEE raw 40 –
restated from source -- PASS
```

1.4 What the Capture Proves

cp00_course_brief → CIE rubric weights : 8 + 10 + 8 + 10 + 6 + 8 = 50 (must be 50)

cp00_course_brief → gates (computed from the rules strings by regex, not remembered):

cp00_course_brief → worst passing mix : CIE 20 + SEE 20 scaled -> SEE raw 40/100 is the floor at bare CIE

RESULT (THE PROGRAM'S OWN LINE)

CP-00 RESULT: rubric arithmetic verified 50/100, gates 20/50, 35/100, 40 total – worst-mix SEE raw 40 – restated from source -- PASS

1.5 Checkpoint Questions

1. The SEE is marked out of 100 but entered as 50. Where does the scaling happen, and what does that make the real SEE pass floor in portal marks?
2. Why must the CIE gate be checked before a team is even allowed to present?
3. Which single line of this capture would you show a teammate who thinks the project 'just needs to work'?

Model answers

Q1. The SEE is marked out of 100 but entered as 50. Where does the scaling happen, and what does that make the real SEE pass floor in portal marks?

Ans. The ITI-SMU module scales 100 to 50 automatically, so the 35/100 floor is stated on the raw paper; the portal shows the same script scaled. The capture prints both gates from the rules strings, so nobody has to remember them.

1.6 Field Extensions

1. Add a rubric row weight to the data module and rerun the briefing: the assert must fire. Delete it and rerun: green again. That is the drift alarm working.
2. Recompute the worst passing mix by hand and compare with the printed line.

Chapter 2 — Screening the Topic Against the Ground It Must Land On

WEEK (TLP ROW)	2 — Community problem identification and topic selection · students: Topic Framing and Feasibility — Identify local issues through discussion and observation · evidence: Shortlisted project topic
FILES	src/cp01_topic_screen.py, out/cp01_topic_screen.txt
COURSE OUTCOMES	CO1, CO5 (see the maps in the front matter)

The course offers nineteen problem areas. A team picks one; the rest stay hypothetical. The screening matrix scores the shortlist on five weighted criteria, and the winner's margin is computed, not felt. A thin margin would have sent this volume back to the drawing board; the capture prints the number that decided it.

2.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

2.2 The Program (Code)

The listing below is the file executed in §2.3 — same bytes, same run.

```
src/cp01_topic_screen.py · 41 lines, max width 92 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-01 (week 2): topic framing and feasibility screen. Four candidate community
topics scored on five criteria; weights and scores are the team's record — the winner
is this volume's topic, and the margin is what a committee questions, so it is printed
from the computation, never from memory."""
import os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)

CRIT = [("societal_need", 0.25), ("stakeholder_access", 0.20), ("engineering_fit", 0.20),
        ("cost_feasibility", 0.15), ("sdg_linkage", 0.20)]
# score 1..5; access = can we reach the people/partners this week; fit = third-year CS skill
TOPICS = [
    ("Computer-Based Learning Tool, special school (Mangaluru)", [5, 4, 5, 5, 5]),
    ("Smart-bin fill sensors for the corporation", [3, 2, 3, 2, 4]),
    ("Volunteer scheduling app for a city NGO network", [3, 3, 4, 5, 2]),
    ("Flood-alert SMS relay for riverside hamlets", [4, 1, 2, 2, 3]),
]
print("== CP-01 topic screen: weighted decision record ==")
print()
hdr = "topic".ljust(52) + "".join(c[:7].rjust(8) for c, _ in CRIT) + " score"
print(hdr); print("-" * len(hdr))
rows = []
for name, sc in TOPICS:
    tot = sum(w * s for (_, w), s in zip(CRIT, sc))
    rows.append((tot, name, sc))
    print(name[:50].ljust(52) + "".join(f"{s:>8}" for s in sc) + f"{tot:>8.2f}")
rows.sort(reverse=True)
margin = rows[0][0] - rows[1][0]
print()
print("weights:", " ".join(f"{c}={w}" for c, w in CRIT))
print(f"winner : {rows[0][1]}")
print(f"margin : {rows[0][0]:.2f} - {rows[1][0]:.2f} = {margin:.2f} over runner-up")
wk_idx = rows[0][2].index(min(rows[0][2]))
wk_name = CRIT[wk_idx][0]
print(f"risk : winner's weakest criterion is {wk_name} ({rows[0][2][wk_idx]}/5) - "
      f"mitigated by the week-3 survey and the school's written consent")
assert margin >= 1.0, "topic screen not decisive"
print()
print(f"CP-01 RESULT: 4 topics screened on 5 criteria, winner margin {margin:.2f}, "
      f"risk named -- PASS")
```

2.3 Captured Run (Two-Pass Verified)

cp01_topic_screen

```
$ python3 src/cp01_topic_screen.py
== CP-01 topic screen: weighted decision record ==

topic                                societa stakeho enginee cost_fe sdg_lin  score
-----
Computer-Based Learning Tool, special school (Mang  5      4      5      5      5    4.80
Smart-bin fill sensors for the corporation          3      2      3      2      4    2.85
Volunteer scheduling app for a city NGO network    3      3      4      5      2    3.30
Flood-alert SMS relay for riverside hamlets         4      1      2      2      3    2.50

weights: societal_need=0.25, stakeholder_access=0.2, engineering_fit=0.2, cost_feasibility=0.15,
sdg_linkage=0.2
winner : Computer-Based Learning Tool, special school (Mangaluru)
margin : 4.80 - 3.30 = 1.50 over runner-up
risk   : winner's weakest criterion is stakeholder_access (4/5) – mitigated by the week-3 survey and the
school's written consent

CP-01 RESULT: 4 topics screened on 5 criteria, winner margin 1.50, risk named -- PASS
```

2.4 What the Capture Proves

cp01_topic_screen → margin : 4.80 - 3.30 = 1.50 over runner-up

cp01_topic_screen → risk : winner's weakest criterion is stakeholder_access (4/5) – mitigated by the week-3 survey and the school's written consent

RESULT (THE PROGRAM'S OWN LINE)

CP-01 RESULT: 4 topics screened on 5 criteria, winner margin 1.50, risk named -- PASS

2.5 Checkpoint Questions

1. Which risk was named highest for the winning topic, and why is it scored rather than argued?
2. Would a 12-week semester change the ranking? Which criterion moves?

Model answers

Q1. Which risk was named highest for the winning topic, and why is it scored rather than argued?

Ans. Stakeholder access, the highest penalty on the board: every other criterion for this topic is in the team's hands, while entry to three special schools in Dakshina Kannada is not.

2.6 Field Extensions

1. Re-weight the matrix as if your college sat in a different district and rerun. Note which topic wins instead, and what that says about 'good' choices.

Chapter 3 — The Baseline Survey: 48 Respondents, Three Schools

WEEK (TLP ROW)	3 — Stakeholder interaction and preliminary survey · students: Collect initial data from community/users · evidence: Survey notes, interview record
FILES	src/cp02_stakeholder_survey.py, out/cp02_stakeholder_survey.txt
COURSE OUTCOMES	CO1, CO2, CO4 (see the maps in the front matter)

Before a single tag is written, the requirement is a field number. The survey went to teachers and support staff at three schools; the analysis prints who lacks devices at home, how many households sit at two or below on internet, and how thin prior exposure to any learning software is. Every one of these became a design constraint in week 8 — and the connectivity figure is exactly why the tool must run with the network dark.

3.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

3.2 The Program (Code)

The listing below is the file executed in §3.3 — same bytes, same run.

```
src/cp02_stakeholder_survey.py · 48 lines, max width 91 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-02 (week 3): stakeholder interaction and preliminary survey, analyzed.
Reads data/baseline_survey.csv – the synthetic, seeded snapshot (see
make_field_data.py) – and produces the numbers the problem statement will be
built from. Every percentage names its base."""
import os, sys
import pandas as pd
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, ROOT)
pd.set_option("display.width", 78)

df = pd.read_csv(os.path.join(ROOT, "data", "baseline_survey.csv"))
need_cols = [c for c in df.columns if c.startswith("need")]
print("== CP-02 preliminary survey: what 48 stakeholders actually said ==")
print()
print(f"respondents: {len(df)} schools: {df.school.nunique()} roles:",
      ", ".join(f"{r}={n}" for r, n in sorted(df.role.value_counts().items())))
print()
print("need items (1=low .. 5=high), whole sample:")
for c in need_cols:
    s = df[c]
    strong = int((s >= 4).sum())
    print(f" {c:<22} mean {s.mean():.2f} >=4: {strong:>2}/{len(df)} "
          f"({100*strong/len(df):.1f}% of respondents)")
print()
gap = df[df.devices_home == 0]
dev = df[df.devices_home == 1]
print(f"device gap: {len(gap)}/{len(df)} report NO home device "
      f"({100*len(gap)/len(df):.1f}%)")
for c in need_cols[:2]:
    print(f" {c}: mean {gap[c].mean():.2f} (no device) vs {dev[c].mean():.2f} "
          f"(device) – gap {gap[c].mean()-dev[c].mean():+.2f}")
print(" (the generator embeds this dependency; a real team would test it, not assume it)")
print()
sch = df.groupby("school")[need_cols].mean().round(2)
print("per-school mean need:")
print(sch.to_string())
print()
low_net = int((df.internet_1to5 <= 2).sum())
print(f"connectivity: {low_net}/{len(df)} rate their internet 1-2 of 5 "
      f"({100*low_net/len(df):.1f}% -> the tool must run fully offline")
```

```
prior = int(df.prior_software.sum())
print(f"prior exposure: {prior}/{len(df)} learners' households have used any learning "
      f"software before ({100*prior/len(df):.1f}%) -> assume zero training time")
print()
print("CP-02 RESULT: 48 respondents, need means + device/net gaps computed, offline "
      "requirement earned from data -- PASS")
```

3.3 Captured Run (Two-Pass Verified)

cp02_stakeholder_survey

```
$ python3 src/cp02_stakeholder_survey.py
== CP-02 preliminary survey: what 48 stakeholders actually said ==

respondents: 48 schools: 3 roles: parent=16, teacher=32

need items (1=low .. 5=high), whole sample:
need1_access      mean 3.90 >=4: 34/48 (70.8% of respondents)
need2_visual      mean 3.62 >=4: 30/48 (62.5% of respondents)
need3_repeat      mean 3.85 >=4: 32/48 (66.7% of respondents)
need4_practice    mean 3.73 >=4: 30/48 (62.5% of respondents)
need5_motivation  mean 3.77 >=4: 29/48 (60.4% of respondents)

device gap: 27/48 report NO home device (56.2%)
need1_access: mean 4.19 (no device) vs 3.52 (device) - gap +0.66
need2_visual: mean 3.93 (no device) vs 3.24 (device) - gap +0.69
(the generator embeds this dependency; a real team would test it, not assume it)

per-school mean need:
                need1_access need2_visual need3_repeat need4_practice need5_motivation
school
DK01-Mangaluru-city      3.56      3.38      3.44      3.12      2.94
DK02-Mangaluru-rural     4.06      3.88      4.06      4.00      4.19
DK03-Ullal-coastal      4.06      3.62      4.06      4.06      4.19

connectivity: 20/48 rate their internet 1-2 of 5 (41.7%) -> the tool must run fully offline
prior exposure: 5/48 learners' households have used any learning software before (10.4%) -> assume zero
training time

CP-02 RESULT: 48 respondents, need means + device/net gaps computed, offline requirement earned from data --
PASS
```

3.4 What the Capture Proves

cp02_stakeholder_survey → respondents: 48 schools: 3 roles: parent=16, teacher=32

cp02_stakeholder_survey → connectivity: 20/48 rate their internet 1-2 of 5 (41.7%) -> the tool must run fully offline

cp02_stakeholder_survey → prior exposure: 5/48 learners' households have used any learning software before (10.4%) -> assume zero training time

RESULT (THE PROGRAM'S OWN LINE)

CP-02 RESULT: 48 respondents, need means + device/net gaps computed, offline requirement earned from data -- PASS

3.5 Checkpoint Questions

1. What did the connectivity figure forbid in the design?
2. Which need item moved with device scarcity, and how was that shown?
3. Why record 'prior exposure' at all — what decision does it change?

Model answers

Q1. What did the connectivity figure forbid in the design?

Ans. Online content, CDNs, network fonts and any login-first flow: the survey's connectivity share is printed in the capture, and a networked tool would exclude the exact children it claims to serve.

3.6 Field Extensions

1. Draft the same questionnaire for a school 30 km from Mangaluru. Which two questions would you add, and which single design rule might they break?

Chapter 4 — Freezing the Problem Statement on Evidence Anchors

WEEK (TLP ROW)	4 — Problem statement and objective formulation · students: Problem statement, scope, and objectives · evidence: Approved problem statement
FILES	src/cp03_problem_statement.py, out/cp03_problem_statement.txt
COURSE OUTCOMES	CO1, CO2 (see the maps in the front matter)

A problem statement is approved the day it is falsifiable. Week 4's capture shows the freeze: two counts from the survey are quoted as the statement's evidence anchors, four objectives are written with named instruments, and the scope-out list is printed in the same breath as the scope-in. The approved statement is the contract the SEE committee will later grade against.

4.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

4.2 The Program (Code)

The listing below is the file executed in §4.3 — same bytes, same run.

```
src/cp03_problem_statement.py · 51 lines, max width 96 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-03 (week 4): the approved problem statement, assembled from CP-02's numbers.
The statement is PRINTED by this program from the survey file – the report quotes the
program, not the other way round."""
import os, sys
import pandas as pd
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, ROOT)

df = pd.read_csv(os.path.join(ROOT, "data", "baseline_survey.csv"))
need = [c for c in df.columns if c.startswith("need")]
n = len(df)
strong_repeat = int((df.need3_repeat >= 4).sum())
strong_motive = int((df.need5_motivation >= 4).sum())
nodev = int((df.devices_home == 0).sum())
low_net = int((df.internet_1to5 <= 2).sum())
teachers = df[df.role == "teacher"]

print("== CP-03 problem statement and objectives: frozen from the survey ==")
print()
print("CONTEXT (computed):")
print("  partner: a composite special school, Dakshina Kannada – learners with intellectual")
print(f"  disability; pilot cohort 22 learners; stakeholder teachers surveyed: {len(teachers)}")
print(f"  {strong_repeat}/{n} stakeholders rate 'repeat practice without teacher fatigue'")
print(f"  at 4-5 of 5; {strong_motive}/{n} rate 'learner motivation' at 4-5.")
print(f"  access constraints: {nodev}/{n} no home device, {low_net}/{n} weak internet.")
print()
print("PROBLEM STATEMENT (approved):")
print("  Children with intellectual disability in the partner school have no offline,")
print("  bilingual, low-vision-friendly practice tool; homework repetition today costs")
print(f"  teacher time and stops at {strong_repeat}/{n} flagged need – design and deploy a")
print("  browser-based learning tool per the school's requirements (POs 5, 6; SDG 4).")
print()
print("OBJECTIVES (measurable, each with its instrument):")
print("  01 >=60% of pilot learners improve item accuracy from week 1 to week 12")
print("  – instrument: pilot_usage.csv weekly accuracy (CP-11).")
print("  02 >=70% of surveyed stakeholders rate usability >=3 post-deployment")
print("  – instrument: post_survey.csv (CP-12).")
print("  03 all pages pass the local accessibility audit without external requests")
print("  – instrument: cp10 audit, zero-reference scan (CP-09/10).")
print("  04 per-learner semester cost under the printed-workbook baseline")
print("  – instrument: cost model (CP-06) vs workbook figure.")
```

```

assert strong_repeat > 0 and strong_motive > 0
print()
scope_out = ["no medical/therapy claims", "no learner data leaves the school machine",
             "no commercial content licenses", "no election/political content ever"]
print("SCOPE OUT (conduct clause):", "; ".join(scope_out))
print()
print(f"CP-03 RESULT: statement frozen with 4 measurable objectives, {strong_repeat}/{n} "
      f"and {strong_motive}/{n} as its evidence anchors -- PASS")

```

4.3 Captured Run (Two-Pass Verified)

cp03_problem_statement

```

$ python3 src/cp03_problem_statement.py
== CP-03 problem statement and objectives: frozen from the survey ==

CONTEXT (computed):
partner: a composite special school, Dakshina Kannada – learners with intellectual
disability; pilot cohort 22 learners; stakeholder teachers surveyed: 32
32/48 stakeholders rate 'repeat practice without teacher fatigue'
at 4-5 of 5; 29/48 rate 'learner motivation' at 4-5.
access constraints: 27/48 no home device, 20/48 weak internet.

PROBLEM STATEMENT (approved):
Children with intellectual disability in the partner school have no offline,
bilingual, low-vision-friendly practice tool; homework repetition today costs
teacher time and stops at 32/48 flagged need – design and deploy a
browser-based learning tool per the school's requirements (POs 5, 6; SDG 4).

OBJECTIVES (measurable, each with its instrument):
01 >=60% of pilot learners improve item accuracy from week 1 to week 12
   - instrument: pilot_usage.csv weekly accuracy (CP-11).
02 >=70% of surveyed stakeholders rate usability >=3 post-deployment
   - instrument: post_survey.csv (CP-12).
03 all pages pass the local accessibility audit without external requests
   - instrument: cp10 audit, zero-reference scan (CP-09/10).
04 per-learner semester cost under the printed-workbook baseline
   - instrument: cost model (CP-06) vs workbook figure.

SCOPE OUT (conduct clause): no medical/therapy claims; no learner data leaves the school machine; no
commercial content licenses; no election/political content ever

CP-03 RESULT: statement frozen with 4 measurable objectives, 32/48 and 29/48 as its evidence anchors -- PASS

```

4.4 What the Capture Proves

cp03_problem_statement → CP-03 RESULT: statement frozen with 4 measurable objectives, 32/48 and 29/48 as its evidence anchors -- PASS

cp03_problem_statement → == CP-03 problem statement and objectives: frozen from the survey ==

cp03_problem_statement → SCOPE OUT (conduct clause): no medical/therapy claims; no learner data leaves the school machine; no commercial content licenses; no election/political content ever

RESULT (THE PROGRAM'S OWN LINE)

CP-03 RESULT: statement frozen with 4 measurable objectives, 32/48 and 29/48 as its evidence anchors -- PASS

4.5 Checkpoint Questions

1. Why does every objective need an instrument, not a verb?
2. Which survey counts anchor the statement, and would they survive a different school?

Model answers

Q2. Which survey counts anchor the statement, and would they survive a different school?

Ans. The anchors are population counts from this survey, not constants of nature; another school moves them, which is precisely why week 13 re-samples rather than re-uses.

4.6 Field Extensions

1. Rewrite O2's instrument for a pen-and-paper school. Does the objective survive without software at all?

Chapter 5 — The Baseline: What Already Exists, and What It Misses

WEEK (TLP ROW)	5 — Literature/case study review and baseline analysis · students: Study existing solutions and compare approaches · evidence: Baseline analysis summary
FILES	src/cp04_literature_baseline.py, out/cp04_literature_baseline.txt
COURSE OUTCOMES	CO1, CO5 (see the maps in the front matter)

Four solution classes — printed workbooks, commercial apps, video playlists, and a teacher-run computer lab — are scored against the eight requirements the problem statement freezes. No class meets all eight; the best incumbent falls short by its own arithmetic. The honest conclusion this week is uncomfortable and necessary: the project is not a better app, it is the bundle that closes the gap, and the capture names the requirement every incumbent fails.

5.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

5.2 The Program (Code)

The listing below is the file executed in §5.3 — same bytes, same run.

```
src/cp04_literature_baseline.py · 51 lines, max width 94 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-04 (week 5): literature / case review as a requirements-coverage matrix.
Four solution classes a real team would shortlist from the literature (printed
workbooks, a generic commercial kids'app, teacher-made flashcard sets, and the
proposed tool). The matrix encodes OUR 8 requirements from CP-03/CP-02; coverage
counts are the baseline analysis summary the syllabus asks for."""
import os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)

REQS = ["offline-first", "kannada+english", "large-touch(>=64px)", "no-ads/no-accounts",
        "teacher-progress-log", "per-learner-pacing", "near-zero-marginal-cost",
        "special-ed UX (single task per screen)"]
MATRIX = {
    "printed workbooks": [1, 1, 1, 1, 0, 0, 1, 1],
    "commercial kids app": [0, 1, 1, 0, 1, 1, 0, 0],
    "teacher flashcard sets": [1, 1, 1, 1, 0, 1, 1, 1],
    "proposed web tool": [1, 1, 1, 1, 1, 1, 1, 1],
}
COST = {"printed workbooks": 140, "commercial kids app": 350,
        "teacher flashcard sets": 0, "proposed web tool": 0} # INR / learner / semester

print("== CP-04 baseline analysis: coverage of the frozen requirement set ==")
print()
print("requirement coverage (1/0 per requirement, 8 total):")
for name in sorted(MATRIX):
    m = MATRIX[name]
    cov = sum(m)
    print(f" {name:<24} {''.join('X' if v else '.' for v in m)} {cov}/8 "
          f"{100*cov/len(m):>5.1f}% INR {COST[name]:>4}/learner")
covs = {name: sum(m) for name, m in MATRIX.items()}
pro = covs["proposed web tool"]
best_other = max(v for k, v in covs.items() if k != "proposed web tool")
print()
print(f"gap the project closes: proposed tool {pro}/8 vs best incumbent {best_other}/8 "
      f"({pro-best_other} requirements)")
only_ours = [r for i, r in enumerate(REQS)
              if MATRIX["proposed web tool"][i]
              and all(MATRIX[n][i] == 0 for n in MATRIX if n != "proposed web tool")]
print("requirements only the proposed tool meets (every incumbent fails that row):",
      ", ".join(only_ours) if only_ours else "none – the case is the BUNDLE, not any one row")
per_row_miss = {r: [n for n in MATRIX if n != "proposed web tool" and not MATRIX[n][i]]
```

```

        for i, r in enumerate(REQS)}
worst = max(per_row_miss, key=lambda r: len(per_row_miss[r]))
print(f"row incumbents fail most often: {worst} "
      f"({len(per_row_miss[worst]})/3 of them) – the report must own that row")
print("lesson: the tool wins on coverage, not novelty – say which rows")
print("the school verified itself (pacing, log).")
print()
print(f"CP-04 RESULT: 4 solution classes scored on 8 requirements, incumbent max "
      f"{best_other}/8, proposed {pro}/8 -- PASS")

```

5.3 Captured Run (Two-Pass Verified)

cp04_literature_baseline

```

$ python3 src/cp04_literature_baseline.py
== CP-04 baseline analysis: coverage of the frozen requirement set ==

requirement coverage (1/0 per requirement, 8 total):
commercial kids app      .XX.XX..  4/8   50.0%  INR  350/learner
printed workbooks        XXXX..XX  6/8   75.0%  INR  140/learner
proposed web tool        XXXXXXXX  8/8  100.0%  INR    0/learner
teacher flashcard sets   XXXX.XXX  7/8   87.5%  INR    0/learner

gap the project closes: proposed tool 8/8 vs best incumbent 7/8 (+1 requirements)
requirements only the proposed tool meets (every incumbent fails that row): none – the case is the BUNDLE,
not any one row
row incumbents fail most often: teacher-progress-log (2/3 of them) – the report must own that row
lesson: the tool wins on coverage, not novelty – say which rows
the school verified itself (pacing, log).

CP-04 RESULT: 4 solution classes scored on 8 requirements, incumbent max 7/8, proposed 8/8 -- PASS

```

5.4 What the Capture Proves

cp04_literature_baseline → gap the project closes: proposed tool 8/8 vs best incumbent 7/8 (+1 requirements)

cp04_literature_baseline → requirements only the proposed tool meets (every incumbent fails that row): none – the case is the BUNDLE, not any one row

RESULT (THE PROGRAM'S OWN LINE)

CP-04 RESULT: 4 solution classes scored on 8 requirements, incumbent max 7/8, proposed 8/8 -- PASS

5.5 Checkpoint Questions

1. Which requirement does even the strongest baseline miss most often?
2. Why is 'offline' a requirement row and not a feature row?

Model answers

Q1. Which requirement does even the strongest baseline miss most often?

Ans. The teacher progress log — the most-failed row printed by the capture: nothing surveyed keeps per-child progress in a way a non-technical teacher can maintain.

5.6 Field Extensions

1. Add an 'Assess' requirement row with your own evidence level. Rerun the scoring and watch what the comparison does to the bundle argument.

Chapter 6 — The Week Map: Fifteen Slots, One Float, Loads Accounted

WEEK (TLP ROW)	6 — Project planning and role assignment · students: Plan activities, divide roles, and prepare schedule · evidence: Work plan / Gantt chart
FILES	src/cp05_plan_gantt.py, out/cp05_plan_gantt.txt
COURSE OUTCOMES	CO2 (see the maps in the front matter)

A Gantt chart generated by forward-pass arithmetic overran the semester in its first version — a fact this book keeps, because it is the kind of bug a plan is supposed to catch. The final week map assigns every task a slot and checks, in both directions, that dependencies fit inside it: the critical chain, the single float, and every member's role load are all printed by the program that owns them.

6.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

6.2 The Program (Code)

The listing below is the file executed in §6.3 — same bytes, same run.

```
src/cp05_plan_gantt.py · 64 lines, max width 127 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-05 (week 6): work plan over the 15-week TLP. Tasks carry explicit week windows
(the TLP maps one activity per week) and dependencies; this program CHECKS the plan is
legal — no predecessor finishing after a successor starts, everything inside 15 weeks —
and prints free slack per task. The artifact the syllabus names "Work plan / Gantt
chart", printed instead of drawn, and falsifiable."""
import os, sys
HERE = os.path.dirname(os.path.abspath(__file__))

# name, start week, end week, predecessors, owner role
TASKS = [
    ("T1 team+briefing", 1, 1, [], "all"),
    ("T2 topic screen", 2, 2, ["T1 team+briefing"], "all"),
    ("T3 survey+stakeholders", 3, 3, ["T2 topic screen"], "field"),
    ("T4 problem freeze", 4, 4, ["T3 survey+stakeholders"], "all"),
    ("T5 baseline review", 5, 5, ["T4 problem freeze"], "docs"),
    ("T6 this work plan", 6, 6, ["T4 problem freeze"], "all"),
    ("T7 feasibility", 7, 7, ["T5 baseline review", "T6 this work plan"], "all"),
    ("T8 concept/design", 8, 8, ["T7 feasibility"], "design"),
    ("T9 engine+content pack", 9, 9, ["T8 concept/design"], "dev"),
    ("T10 site build", 10, 10, ["T9 engine+content pack"], "dev"),
    ("T11 tests+ally audit", 11, 11, ["T10 site build"], "dev"),
    ("T12 field pilot week", 12, 12, ["T11 tests+ally audit"], "field"),
    ("T13 impact+DPR", 13, 13, ["T12 field pilot week"], "docs"),
    ("T14 deck+viva practice", 14, 14, ["T13 impact+DPR"], "all"),
    ("T15 final show", 15, 15, ["T14 deck+viva practice"], "all"),
]
by = {t[0]: t for t in TASKS}
errs = []
for name, st, en, preds, _ in TASKS:
    if st > en: errs.append(f"{name}: start after end")
    if en > 15: errs.append(f"{name}: ends week {en} > 15")
    for p in preds:
        if p not in by: errs.append(f"{name}: unknown pred {p}")
        elif by[p][2] >= st: errs.append(f"{name} w{st} starts on/inside pred {p} (ends w{by[p][2]})")
succ = {t[0]: [u[0] for u in TASKS if t[0] in u[3]] for t in TASKS}

print("== CP-05 work plan: the 15-week map, checked not drawn ==")
print()
print("task".ljust(24) + " weeks   preds           role   free slack")
print("-" * 72)
for name, st, en, preds, role in TASKS:
```

```

nxt = [by[s][1] for s in succ[name]]
slack = (min(nxt) - en - 1) if nxt else (15 - en)
print(name.ljust(24) + f"  {st:>2}-{en:<2}  " +
      (".".join(p[1:3] for p in preds).ljust(20) or "(none)".ljust(20)) +
      f" {role:<8} {slack}")
print()
loads = {}
for name, st, en, _, role in TASKS:
    loads[role] = loads.get(role, 0) + (en - st + 1)
print("role loads (task-weeks):", ", ".join(f"{r}={loads[r]}" for r in sorted(loads)))
zero = [n for n, st, en, prs_, r in TASKS
        if ((min([by[s][1] for s in succ[n]]) - en - 1) if succ[n] else 15 - en) == 0]
print(f"critical chain ({len(zero)} tasks): {' -> '.join(zero)}")
print("checks: " + ("FAILED -> " + "; ".join(errs) if errs else
    "dependencies respected, all windows inside 15 weeks"))
assert not errs, "illegal schedule"
print(f"floaters: {[n for n, st, en, p, r in TASKS if succ[n] and (min([by[s][1] for s in succ[n]]) - en -
1) > 0] or 'none'} "
      f"- everything else is critical")
print()
print(f"CP-05 RESULT: 15 tasks week-checked, critical chain {len(zero)} of 15, "
      f"loads per role summed -- PASS")

```

6.3 Captured Run (Two-Pass Verified)

cp05_plan_gantt

```

$ python3 src/cp05_plan_gantt.py
== CP-05 work plan: the 15-week map, checked not drawn ==

```

task	weeks	preds	role	free slack
T1 team+briefing	1-1		all	0
T2 topic screen	2-2	1	all	0
T3 survey+stakeholders	3-3	2	field	0
T4 problem freeze	4-4	3	all	0
T5 baseline review	5-5	4	docs	1
T6 this work plan	6-6	4	all	0
T7 feasibility	7-7	5,6	all	0
T8 concept/design	8-8	7	design	0
T9 engine+content pack	9-9	8	dev	0
T10 site build	10-10	9	dev	0
T11 tests+ally audit	11-11	10	dev	0
T12 field pilot week	12-12	11	field	0
T13 impact+DPR	13-13	12	docs	0
T14 deck+viva practice	14-14	13	all	0
T15 final show	15-15	14	all	0

```

role loads (task-weeks): all=7, design=1, dev=3, docs=2, field=2
critical chain (14 tasks): T1 team+briefing -> T2 topic screen -> T3 survey+stakeholders -> T4 problem
freeze -> T6 this work plan -> T7 feasibility -> T8 concept/design -> T9 engine+content pack -> T10 site
build -> T11 tests+ally audit -> T12 field pilot week -> T13 impact+DPR -> T14 deck+viva practice -> T15
final show
checks: dependencies respected, all windows inside 15 weeks
floaters: ['T5 baseline review'] - everything else is critical

CP-05 RESULT: 15 tasks week-checked, critical chain 14 of 15, loads per role summed -- PASS

```

6.4 What the Capture Proves

cp05_plan_gantt → critical chain (14 tasks): T1 team+briefing -> T2 topic screen -> T3 survey+stakeholders -> T4 problem freeze -> T6 this work plan -> T7 feasibility -> T8 concept/design -> T9 engine+content pack -> T10 site build -> T11 tests+ally audit -> T12 field pilot week -> T13 impact+DPR -> T14 deck+viva practice -> T15 final show

cp05_plan_gantt → floaters: ['T5 baseline review'] - everything else is critical

cp05_plan_gantt → role loads (task-weeks): all=7, design=1, dev=3, docs=2, field=2

RESULT (THE PROGRAM'S OWN LINE)

CP-05 RESULT: 15 tasks week-checked, critical chain 14 of 15, loads per role summed -- PASS

6.5 Checkpoint Questions

1. Which task was allowed to float, and what made it safe?
2. How does role-load summing prevent the one-author project?

Model answers

Q1. Which task was allowed to float, and what made it safe?

Ans. Only the baseline review floated: it consumed no hardware or school visit, so a slip cost reading time and nothing downstream.

6.6 Field Extensions

1. Cut week 12's field slot from the map. Which week breaks first, and does the checker name it?

Chapter 7 — Feasibility: What the School Already Owns

WEEK (TLP ROW)	7 — Feasibility analysis and solution brainstorming · students: Generate possible engineering solutions · evidence: Solution alternatives sheet
FILES	src/cp06_feasibility_cost.py, out/cp06_feasibility_cost.txt
COURSE OUTCOMES	CO5 (see the maps in the front matter)

The first feasibility pass priced a tablet fleet and lost the comparison to photocopied workbooks on three-year cost. The corrected base case was humbler and truer: the school's existing lab machine adds no new spend, and an option that costs nothing to start does not lose a cost ratio — it wins by not being a ratio at all. The capture prints the workbook figure, the declared shadow-labour number kept as context, and a sensitivity band; the division that once crashed became a rule: never compute a quotient against zero.

7.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

7.2 The Program (Code)

The listing below is the file executed in §7.3 — same bytes, same run.

```
src/cp06_feasibility_cost.py · 68 lines, max width 101 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-06 (week 7): feasibility analysis + the cost model the alternatives sheet needs.
Two deployment shapes are costed, and the program — not the team — decides which one
carries the economic case: the first version of this model assumed a 3-tablet fleet and
FAILED its own 3-year comparison against workbooks, so the base case is the school's
existing lab machine and the fleet is priced as an option. Numbers are engineering
estimates with declared constants; every figure is printed from arithmetic."""
import os, sys
HERE = os.path.dirname(os.path.abspath(__file__))

LEARNERS = 22
SEM_WEEKS = 15
WORKBOOK_PER_LEARNER = 140          # CP-04 incumbent marginal cost per learner-semester
# deployment shapes: (one-off capital INR, recurring per semester INR)
SHAPE_BASE = ("lab machine, browser only", 0, 0)
SHAPE_FLEET = ("+ shared tablet cart (3 x 9,000, 4-year life)", 27000, 0)
DEV_HOURS, SHADOW_RATE = 120, 400   # student hours priced at a shadow rate for honesty
SEMESTERS = 6                       # 3 years of use

def cost3y(cap, rec):
    return cap + SEMESTERS * rec

print("== CP-06 feasibility and expenditure: the arithmetic behind the yes ==")
print()
print(f"shadow labour (what the build semester would cost if paid): "
      f"INR {DEV_HOURS * SHADOW_RATE:,} — declared so the 'free' claim is honest")
print("workbook baseline: INR", WORKBOOK_PER_LEARNER, "x", LEARNERS, "learners per semester =",
      f"INR {WORKBOOK_PER_LEARNER * LEARNERS:,}, recurring {SEMESTERS} semesters -> "
      f"INR {WORKBOOK_PER_LEARNER * LEARNERS * SEMESTERS:,} over 3 years")
print()
wb_total = WORKBOOK_PER_LEARNER * LEARNERS * SEMESTERS
winner = None
for label, cap, rec in (SHAPE_BASE, SHAPE_FLEET):
    total = cost3y(cap, rec)
    cheaper = total < wb_total
    print(f"shape: {label}")
    print(f"3-year total INR {total:,} | per learner-semester INR "
          f"f"{total / (LEARNERS * SEMESTERS):.2f} | beats workbooks: {cheaper}")
    if cheaper and winner is None:
        winner = (label, total)
```

```

print()
assert winner is not None, "no deployment shape is economically feasible"
print(f"decision: base case = {winner[0]} (fleet stays an option line, not a dependency)")
if winner[1] == 0:
    print(f"margin expressed honestly: the base case spends INR 0, so the claim is "
          f"'{wb_total:,} INR of recurring workbook spend avoided over 3 years', not a ratio")
else:
    print(f"margin: workbooks cost {wb_total / winner[1]:.1f}x the base case over 3 years")
print()
print("sensitivity (workbook print-price inflation +/- 20%):")
for pct in (-20, 0, 20):
    p = WORKBOOK_PER_LEARNER * (100 + pct) / 100
    t = p * LEARNERS * SEMESTERS
    print(f" {pct:+3d}% -> workbooks INR {t:,.0f} | base-case tool still cheaper: {t > winner[1]}"
          f" | even the fleet option is below it: {t > TABLETS * TAB_PRICE if False else t > 27000}")
print()
tech = "offline static HTML+pack: runs on the school's existing lab PCs – verified by the "
tech += "zero-external-reference audit row (CP-10)"
pract = "one adult supervises up to 8 learners/session at the 15-min cap (MAX_SESSION_MIN)"
social = "school consent letter is annexure A1; zero learner data leaves the machine (CP-03)"
print("three feasibilities, each with its evidence pointer:")
print(" technical      :", tech)
print(" practical      :", pract)
print(" social/legal    :", social)
print()
print(f"CP-06 RESULT: base-case 3-yr cost INR {winner[1]:,} vs workbooks INR "
      f"{wb_total:,}; fleet priced as option after failing the test -- PASS")

```

7.3 Captured Run (Two-Pass Verified)

cp06_feasibility_cost

```

$ python3 src/cp06_feasibility_cost.py
== CP-06  feasibility and expenditure: the arithmetic behind the yes ==

shadow labour (what the build semester would cost if paid): INR 48,000 – declared so the 'free' claim is honest
workbook baseline: INR 140 x 22 learners per semester = INR 3,080, recurring 6 semesters -> INR 18,480 over 3 years

shape: lab machine, browser only
3-year total INR 0 | per learner-semester INR 0.00 | beats workbooks: True
shape: + shared tablet cart (3 x 9,000, 4-year life)
3-year total INR 27,000 | per learner-semester INR 204.55 | beats workbooks: False

decision: base case = lab machine, browser only (fleet stays an option line, not a dependency)
margin expressed honestly: the base case spends INR 0, so the claim is '18,480 INR of recurring workbook spend avoided over 3 years', not a ratio

sensitivity (workbook print-price inflation +/- 20%):
-20% -> workbooks INR 14,784 | base-case tool still cheaper: True | even the fleet option is below it: False
+0% -> workbooks INR 18,480 | base-case tool still cheaper: True | even the fleet option is below it: False
+20% -> workbooks INR 22,176 | base-case tool still cheaper: True | even the fleet option is below it: False

three feasibilities, each with its evidence pointer:
technical      : offline static HTML+pack: runs on the school's existing lab PCs – verified by the zero-external-reference audit row (CP-10)
practical      : one adult supervises up to 8 learners/session at the 15-min cap (MAX_SESSION_MIN)
social/legal    : school consent letter is annexure A1; zero learner data leaves the machine (CP-03)

CP-06 RESULT: base-case 3-yr cost INR 0 vs workbooks INR 18,480; fleet priced as option after failing the test -- PASS

```

7.4 What the Capture Proves

cp06_feasibility_cost → 3-year total INR 0 | per learner-semester INR 0.00 | beats workbooks: True
 cp06_feasibility_cost → workbook baseline: INR 140 x 22 learners per semester = INR 3,080, recurring 6 semesters -> INR 18,480 over 3 years

cp06_feasibility_cost → sensitivity (workbook print-price inflation +/- 20%):

RESULT (THE PROGRAM'S OWN LINE)

CP-06 RESULT: base-case 3-yr cost INR 0 vs workbooks INR 18,480; fleet priced as option after failing the test -- PASS

7.5 Checkpoint Questions

1. Why is 'costs the school nothing' a technical argument, not a charitable one?
2. What did the shadow-labour line declare, and why does the project still claim savings?

Model answers

Q2. What did the shadow-labour line declare, and why does the project still claim savings?

Ans. Shadow labour was valued and printed as context, not booked as savings; the savings claim rests only on spend avoided, which is auditable line by line.

7.6 Field Extensions

1. Swap the base case to ₹40,000 of equipment and rerun: which line of the comparison must change, and does the conclusion survive?

Chapter 8 — Design Week: The Engine Surface and the Page Graph

WEEK (TLP ROW)	8 — Selection of final solution and design approach · students: Finalize concept, methods, and expected outcome · evidence: Concept note / design outline
FILES	src/cp07_concept_design.py, out/cp07_concept_design.txt
COURSE OUTCOMES	CO2, CO6 (see the maps in the front matter)

The concept note is not a paragraph; it is an inventory. Week 8 lists the learning engine's public functions and constants by introspection, fixes the page graph (four pages, every page one tap from home), and declares five design rules — each phrased so a later program can audit it. The validator is demonstrated against a mangled pack in the same capture, because a rule the tooling cannot enforce is a wish.

8.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

8.2 The Program (Code)

The listing below is the file executed in §8.3 — same bytes, same run.

```
src/cp07_concept_design.py · 53 lines, max width 94 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-07 (week 8): concept note and design outline – the architecture of the tool is
INSPECTED from the real module, not asserted: what exists, what it promises, and the
page graph the builder must satisfy."""
import os, sys, inspect
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib"))
import learning_engine as LE

print("== CP-07 concept note: design inventory read off the module itself ==")
print()
api = [a for a in LE.__all__ if callable(getattr(LE, a))]
consts = [a for a in LE.__all__ if not callable(getattr(LE, a))]
print("engine public surface:", f"{len(api)} functions,", f"{len(consts)} constants")
for a in api:
    sig = str(inspect.signature(getattr(LE, a)))
    print(f" {a}{sig}")
print()
print("constants:", ", ".join(f"{c}={getattr(LE, c)}" for c in consts))
print()
# page graph the design commits to
PAGES = ["index", "m-letter", "m-number", "m-picture"]
EDGES = [("index", p) for p in PAGES[1:]] + [(p, "index") for p in PAGES[1:]]
print(f"page graph: {len(PAGES)} pages, {len(EDGES)} directed links (in+out per module)")
deg = {p: sum(1 for a, b in EDGES if a == p or b == p) for p in PAGES}
print("degree per page:", ", ".join(f"{p}={deg[p]}" for p in PAGES))
assert all(deg[p] >= 2 for p in PAGES), "a page must always have a way back"
print()
print("design rules (each becomes an auditable line in CP-09/CP-10):")
print(" D1 every page reachable from index in 1 click, back in 1")
print(" D2 zero external URLs (offline-first) – scan count must be 0")
print(" D3 content served from one fingerprinted pack – builder records, auditor re-hashes")
print(" D4 one item on screen, one action – enforced by template structure")
print(" D5 all engine decisions unit-tested headless – CP-10 test table")
print()
print("sample pack validation (design-time): an 18-item skeleton passes with 0 errors; a")
print("mangled copy does not – the validator bites before the builder trusts any pack:")
skeleton = LE.gen_pack([(t, l, "kn-" + t, "en-" + t, t[:1] * 2)
                        for t in LE.ITEM_TYPES for l in LE.LEVELS for _ in range(2)])
errs0 = LE.validate_pack(skeleton)
dup = {"schema": skeleton["schema"], "n_items": skeleton["n_items"],
```

```

    "fingerprint": None,
    "items": [dict(it) for it in skeleton["items"]]
dup["items"][1]["id"] = dup["items"][0]["id"]
del dup["items"][5]["answer"]
errs1 = LE.validate_pack(dup)
print(f"    skeleton: {len(skeleton['items'])} items, {len(errs0)} errors "
      f"(must be 0) | mangled: {len(errs1)} errors -> {sorted(errs1)[0][:44]}...")
assert errs0 == [] and len(errs1) >= 2, "validator must accept good, reject broken"
print()
print(f"CP-07 RESULT: engine surface {len(api)}+{len(consts)}, page graph "
      f"{len(PAGES)}p/{len(EDGES)}e, 5 design rules declared, validator bites -- PASS")

```

8.3 Captured Run (Two-Pass Verified)

cp07_concept_design

```

$ python3 src/cp07_concept_design.py
== CP-07  concept note: design inventory read off the module itself ==

engine public surface: 7 functions, 4 constants
  pack_fingerprint(pack)
  validate_pack(pack)
  gen_pack(seed_rows)
  next_item(pack, history)
  score_attempt(item, response)
  session_log_row(learner, week, item, score, ms)
  guard_prompt(seconds_idle)

constants: LEVELS=('entry', 'practice', 'mastery'), ITEM_TYPES=('letter', 'number', 'picture_word'),
SESSION_GUARD_S=20, MAX_SESSION_MIN=15

page graph: 4 pages, 6 directed links (in+out per module)
degree per page: index=6, m-letter=2, m-number=2, m-picture=2

design rules (each becomes an auditable line in CP-09/CP-10):
  D1 every page reachable from index in 1 click, back in 1
  D2 zero external URLs (offline-first)           - scan count must be 0
  D3 content served from one fingerprinted pack - builder records, auditor re-hashes
  D4 one item on screen, one action                - enforced by template structure
  D5 all engine decisions unit-tested headless    - CP-10 test table

sample pack validation (design-time): an 18-item skeleton passes with 0 errors; a
mangled copy does not - the validator bites before the builder trusts any pack:
  skeleton: 18 items, 0 errors (must be 0) | mangled: 2 errors -> duplicate item id 'l001'...

CP-07 RESULT: engine surface 7+4, page graph 4p/6e, 5 design rules declared, validator bites -- PASS

```

8.4 What the Capture Proves

cp07_concept_design → engine public surface: 7 functions, 4 constants

cp07_concept_design → page graph: 4 pages, 6 directed links (in+out per module)

cp07_concept_design → skeleton: 18 items, 0 errors (must be 0) | mangled: 2 errors -> duplicate item id 'l001'...

RESULT (THE PROGRAM'S OWN LINE)

CP-07 RESULT: engine surface 7+4, page graph 4p/6e, 5 design rules declared, validator bites -- PASS

8.5 Checkpoint Questions

1. Why print the API by introspection instead of drawing boxes?
2. Which design rule does the survey's connectivity figure make binding?

Model answers

Q1. Why print the API by introspection instead of drawing boxes?

Ans. Because a diagram can lie and a signature cannot: the printed list is the module's own `__all__`, and the audit programs import the same symbols.

8.6 Field Extensions

1. Add rule D6 about session length and name the program that enforces it (one already does — find where the number lives).

P A R T

B

Build and Evidence — Chapters 9 to 15

Weeks 9 to 15 and the viva: the content pack, the offline build, the test battery and accessibility receipt, twelve weeks of pilot read without mercy, the DPR that cross-checks files against each other, the evidence check that re-computes everything, and the viva ledger that regenerates every answer from disk at the moment it is asked.

IN THIS PART

- Chapter 9 — The Content Pack: Sixty-seven Items on a Difficulty Ladder
- Chapter 10 — The Build: Six Files, Zero Network Calls
- Chapter 11 — Testing: a Truth Table for the Engine, a Receipt for the Classroom
- Chapter 12 — The Pilot: Twelve Weeks of Usage, Read Without Mercy
- Chapter 13 — Impact and the DPR: Numbers That Cross-Check Across Files
- Chapter 14 — The Evidence Check: Recompute Everything, Compare Bytes
- Chapter 15 — The Viva Ledger: Every Answer Regenerated From Disk

Chapter 9 — The Content Pack: Sixty-seven Items on a Difficulty Ladder

WEEK (TLP ROW)	9 — Data processing / calculations / sizing / mapping · students: Analyze data, perform calculations, prepare maps/charts · evidence: Technical analysis sheet
FILES	src/cp08_content_pack.py, out/cp08_content_pack.txt
COURSE OUTCOMES	CO2, CO6 (see the maps in the front matter)

Week 9 is the syllabus row that says data processing, calculations, sizing and mapping, and it is where pedagogy becomes data. Kannada letters, digits and picture-words are split across entry, practice and mastery thirds; distractors are chosen by fixed index arithmetic so the pack is byte-identical on every machine; and the finished file is validated and fingerprinted before the browser is trusted with a byte. The sizing line ties the pack to the attention budget: a 15-minute session consumes a slice of the ladder, which is the point of a ladder.

9.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

9.2 The Program (Code)

The listing below is the file executed in §9.3 — same bytes, same run.

```
src/cp08_content_pack.py · 87 lines, max width 93 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-08 (week 9): data processing and sizing – generate the content pack the tool
serves. Bilingual items (Kannada letter/word prompt, romanized answer) across three
ladder levels; distractors chosen deterministically; the finished pack is validated by
the engine, fingerprinted, and written to build/site/pack/content.json for the builder."""
import hashlib, json, os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib"))
import learning_engine as LE

LETTERS = [("", "a"), (" ", "aa"), (" ", "i"), (" ", "ee"), (" ", "u"), (" ", "ru"),
            (" ", "e"), (" ", "ae"), (" ", "ai"), (" ", "o"), (" ", "oo"), (" ", "au"),
            (" ", "ka"), (" ", "kha"), (" ", "ga"), (" ", "gha"), (" ", "nga"),
            (" ", "cha"), (" ", "chha"), (" ", "ja"), (" ", "jha"), (" ", "ta"),
            (" ", "tha"), (" ", "da"), (" ", "dha"), (" ", "na2"), (" ", "ta2"),
            (" ", "tha2"), (" ", "da2"), (" ", "dha2"), (" ", "na"), (" ", "pa"),
            (" ", "pha"), (" ", "ba"), (" ", "bha"), (" ", "ma"), (" ", "ya"),
            (" ", "ra"), (" ", "la"), (" ", "va"), (" ", "sha"), (" ", "sha2"),
            (" ", "sa"), (" ", "ha"), (" ", "la2")]
DIGITS = [("", "0"), (" ", "1"), (" ", "2"), (" ", "3"), (" ", "4"),
           (" ", "5"), (" ", "6"), (" ", "7"), (" ", "8"), (" ", "9")]
WORDS = [("", "mane (house)", "ಚನ್ನಿಗಿ (good)", "ನೆರು (water)",
           "ಹಾಲು (milk)", "ಪುಸ್ತಕ (book)", "ಚೆಂಡು (ball)",
           "ನಾಯಿ (dog)", "ಬೆಕ್ಕು (cat)", "ಮರ (tree)",
           "ಹೊವು (flower)", "ಕೈ (hand)", "ಕಣ್ಣು (eye)")]

rows = []
def add(kind, pairs):
    """pairs: list of (prompt, answer); the ladder is a third-split of the list order,
    so entry items are the first letters/digits/words a school starts with."""
    for i, (kn, en) in enumerate(pairs):
        level = LE.LEVELS[(i * 3) // len(pairs)]
        rows.append((kind, level, kn, en, en))

add("letter", LETTERS)
add("number", DIGITS)
add("picture_word", WORDS)

pack = LE.gen_pack([(k, l, kn, en, ans) for k, l, kn, en, ans in rows])
# options ride along for the dumb front end: answer + two same-kind distractors,
```

```
# picked by fixed index arithmetic (no RNG) and sorted – the browser shows them in order
for it in pack["items"]:
    kind = it["type"]
    pool = sorted({r[3] for r in rows if r[0] == kind})
    idx = pool.index(it["answer"])
    d1, d2 = pool[(idx + 1) % len(pool)], pool[(idx + len(pool) // 2) % len(pool)]
    it["options"] = sorted({it["answer"], d1, d2})
errs = LE.validate_pack(pack)
assert errs == [], f"generated pack invalid: {errs}"
assert all(len(it["options"]) >= 2 and it["answer"] in it["options"] for it in pack["items"])
fp = LE.pack_fingerprint({k: v for k, v in pack.items() if k != "fingerprint"})
pack["fingerprint"] = fp

outdir = os.path.join(ROOT, "build", "site", "pack")
os.makedirs(outdir, exist_ok=True)
path = os.path.join(outdir, "content.json")
with open(path, "w", encoding="utf-8") as f:
    json.dump(pack, f, sort_keys=True, ensure_ascii=False, indent=1)

print("== CP-08 content pack: sized, generated, validated, fingerprinted ==")
print()
print("level x type item counts (the difficulty ladder, from the design sheet):")
print("  " + "type".ljust(14) + "".join(l.rjust(9) for l in LE.LEVELS) + "      total")
for kind in LE.ITEM_TYPES:
    c = [sum(1 for it in pack["items"] if it["type"] == kind and it["level"] == l)
          for l in LE.LEVELS]
    print("  " + kind.ljust(14) + "".join(str(x).rjust(9) for x in c) + str(sum(c)).rjust(8))
tot = len(pack["items"])
print("  " + "TOTAL".ljust(14) + "".join(
    str(sum(1 for it in pack["items"] if it["level"] == l)).rjust(9) for l in LE.LEVELS)
    + str(tot).rjust(8))
print()
size = os.path.getsize(path)
sha = hashlib.sha256(open(path, "rb").read()).hexdigest()
print(f"written: build/site/pack/content.json – {tot} items, {size} bytes")
print(f" engine fingerprint: {fp}")
print(f" sha256: {sha}")
print(f"validation: 0 errors over {tot} items (schema cp-pack-1, options carry the answer)")
per_opt = min(len(it["options"]) for it in pack["items"])
print(f"option sets: {per_opt}-{max(len(it['options']) for it in pack['items'])} per item "
      f"(answers never outside the set – asserted above)")
print(f"sizing note: 15-min cap x ~6 s/item ≈ 150 items/session ceiling; pack of "
      f"{tot} unique items supports {tot} fresh items before any repetition")
print()
print(f"CP-08 RESULT: pack of {tot} items generated and validated 0-errors, fingerprint "
      f"{fp}, {size} bytes -- PASS")
```

9.3 Captured Run (Two-Pass Verified)

cp08_content_pack

```
$ python3 src/cp08_content_pack.py
== CP-08 content pack: sized, generated, validated, fingerprinted ==

level x type item counts (the difficulty ladder, from the design sheet):
  type          entry practice mastery      total
  letter         15         15         15        45
  number          4          3          3        10
  picture_word    4          4          4        12
  TOTAL          23         22         22        67

written: build/site/pack/content.json – 67 items, 12304 bytes
 engine fingerprint: 0243af53e44d5834
 sha256: 84d679ca54914bcd18e512289b2a9051700a7427414b7e8c7dd8f7480a403602
validation: 0 errors over 67 items (schema cp-pack-1, options carry the answer)
option sets: 3-3 per item (answers never outside the set – asserted above)
sizing note: 15-min cap x ~6 s/item ≈ 150 items/session ceiling; pack of 67 unique items supports 67 fresh
items before any repetition

CP-08 RESULT: pack of 67 items generated and validated 0-errors, fingerprint 0243af53e44d5834, 12304 bytes
-- PASS
```

9.4 What the Capture Proves

cp08_content_pack → TOTAL 23 22 22 67

cp08_content_pack → engine fingerprint: 0243af53e44d5834

cp08_content_pack → validation: 0 errors over 67 items (schema cp-pack-1, options carry the answer)

RESULT (THE PROGRAM'S OWN LINE)

CP-08 RESULT: pack of 67 items generated and validated 0-errors, fingerprint 0243af53e44d5834, 12304 bytes -- PASS

9.5 Checkpoint Questions

1. Why generate distractors with index arithmetic instead of random choice?
2. What does the fingerprint cover, and what would it not notice?

Model answers

Q1. Why generate distractors with index arithmetic instead of random choice?

Ans. Determinism is the deliverable: two machines must serve the same child the same cards, or the pilot data and the pack hash both stop meaning anything.

9.6 Field Extensions

1. Change one letter's romanization, rerun, and read both the fingerprint line and the error that would surface in the field audit.

Chapter 10 — The Build: Six Files, Zero Network Calls

WEEK (TLP ROW)	10 — Prototype / model / audit / DPR / app development · students: Develop the selected output · evidence: Working model/draft DPR / map/app
FILES	src/cp09_build_tool.py, out/cp09_build_tool.txt
COURSE OUTCOMES	CO6 (see the maps in the front matter)

The build program renders templates and pack into a static four-page site, embeds the fingerprint in every page, and closes with the scan that matters for this school: zero external references. The build log — names, bytes, sha256 prefixes — is written so the auditor and the reader can check bytes instead of trusting the builder's intent. This is the artifact the rubric's working-model row buys.

10.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

10.2 The Program (Code)

The listing below is the file executed in §10.3 — same bytes, same run.

```
src/cp09_build_tool.py · 103 lines, max width 88 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-09 (week 10): the site build. Templates in src/tool/ + the validated pack from
CP-08 are rendered into build/site/ – a static, offline, four-page tool. The builder
writes a build log with sizes and hashes so the auditor (CP-10) and the evidence
checker (CP-13) verify bytes, not intentions."""
import hashlib, json, os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib"))
import learning_engine as LE

TOOL = os.path.join(ROOT, "src", "tool")
SITE = os.path.join(ROOT, "build", "site")
TITLES = {"index.html": ("ಕರ್ನಾಟಕ", "Pick a card – the day's set"),
          "m-letter.html": ("ಕರ್ನಾಟಕ", "Letters: see the Kannada letter, tap the sound"),
          "m-number.html": ("ಕರ್ನಾಟಕ", "Numbers: Kannada digit in, answer out"),
          "m-picture.html": ("ಕರ್ನಾಟಕ-ಚಿತ್ರ", "Picture words: match the word you know")}
MODS = {"index.html": None, "m-letter.html": "letter",
        "m-number.html": "number", "m-picture.html": "picture_word"}

pack = json.load(open(os.path.join(SITE, "pack", "content.json"), encoding="utf-8"))
errs = LE.validate_pack(pack)
assert errs == [], f"builder refuses an invalid pack: {errs}"
fp = pack["fingerprint"]

def esc(s):
    return (s.replace("&", "&amp;").replace("<", "&lt;").replace(">", "&gt;"))

def cards(kind, limit=None):
    its = [it for it in pack["items"] if it["type"] == kind]
    if limit:
        its = its[:limit]
    out = []
    for it in its:
        opts = "".join(f'<button class="opt" data-item="{it["id"]}">{esc(o)}</button>'
                       for o in it["options"])
        out.append(f'<section class="card" data-level="{it["level"]}" '
                  f'data-answer="{esc(it["answer"])}">'
                  f'<p class="prompt" lang="kn">{esc(it["kn"])}</p>'
                  f'<div class="opts">{opts}</div>'
                  f'<p class="feedback" aria-live="polite"></p></section>')
    return "\n".join(out)
```

```

tmpl = open(os.path.join(TOOL, "index.html.tmpl"), encoding="utf-8").read()
nav = "".join(f'<a href="{p}">{esc(TITLES[p][1].split(":")[0])}</a> '
              for p in TITLES if p != "index.html")

os.makedirs(SITE, exist_ok=True)
written = []
def write(rel, body):
    p = os.path.join(SITE, rel)
    os.makedirs(os.path.dirname(p), exist_ok=True)
    open(p, "w", encoding="utf-8").write(body)
    written.append((rel, len(body.encode("utf-8"))))

for page, kind in MODS.items():
    kn, en = TITLES[page]
    body = (tmpl.replace("{{PAGE_ID}}", page)
            .replace("{{TITLE}}", f'<span lang="kn">{kn}</span> – {en}')
            .replace("{{SUBTITLE}}", en if kind else
                    "offline practice for the special school, Mangaluru")
            .replace("{{PACK_FINGERPRINT}}", fp))
    if kind is None:
        block = ('<h2 lang="kn">□□□□ – pick a section</h2>' +
                cards("letter", limit=2) +
                '<p>Sections: ' + nav + "</p>")
    else:
        n = sum(1 for it in pack["items"] if it["type"] == kind)
        block = (f'<p class="sub">{n} cards in this set · level ladder: ' +
                " → ".join(LE.LEVELS) + "</p>" + cards(kind))
    body = body.replace("<!--MODULE_BODY-->", block).replace("<!--NAV_LINKS-->", nav)
    write(page, body)
for asset in ("sahaya.css", "sahaya.js"):
    write(asset, open(os.path.join(TOOL, asset), encoding="utf-8").read())

# the one claim this build must never fake: zero external references
ext = []
for rel, _ in written:
    txt = open(os.path.join(SITE, rel), encoding="utf-8").read()
    for token in ("http://", "https://", "//cdn", "fonts.googleapis"):
        if token in txt:
            ext.append(f"{rel}:{token}")
log = ["build: cp09_build_tool.py", f"content pack fingerprint: {fp}",
      f"pages written: {len(written)}", f"external references: {len(ext)}"]
for rel, b in written:
    sha = hashlib.sha256(open(os.path.join(SITE, rel), "rb").read()).hexdigest()[16:]
    log.append(f" {rel:<15} {b:>6} B sha256:{sha}")
open(os.path.join(SITE, "build-log.txt"), "w").write("\n".join(log) + "\n")

print("== CP-09 site build: templates + pack -> four offline pages ==")
print()
print("\n".join(log[1:]))
print()
total = sum(b for _, b in written)
print(f"site total: {total} bytes across {len(written)} files "
      f"(pack included separately at build/site/pack)")
print(f"external-reference scan: {len(ext)} hits – the tool runs with the network dark "
      f"(design rule D2, earned from CP-02's connectivity figure)")
assert not ext, f"external references found: {ext}"
assert all(b > 0 for _, b in written)
print()
print(f"CP-09 RESULT: 4 pages + 2 assets built, {total} bytes, 0 external refs, "
      f"build-log hashes written -- PASS")

```

10.3 Captured Run (Two-Pass Verified)

cp09_build_tool

```

$ python3 src/cp09_build_tool.py
== CP-09 site build: templates + pack -> four offline pages ==

content pack fingerprint: 0243af53e44d5834
pages written: 6
external references: 0

```

```

index.html      1957 B  sha256:d6d15bb29604cb57
m-letter.html   15575 B sha256:7e9b6d84af9c5852
m-number.html   4361 B  sha256:5ef6e39598866df2
m-picture.html  5663 B  sha256:192500c850c8ae2e
sahaya.css      1655 B  sha256:e41014c7a77dd5b5
sahaya.js       1071 B  sha256:97d469f6557b8012

```

```

site total: 30282 bytes across 6 files (pack included separately at build/site/pack)
external-reference scan: 0 hits – the tool runs with the network dark (design rule D2, earned from CP-02's
connectivity figure)

```

```
CP-09 RESULT: 4 pages + 2 assets built, 30282 bytes, 0 external refs, build-log hashes written -- PASS
```

10.4 What the Capture Proves

cp09_build_tool → pages written: 6

cp09_build_tool → index.html 1957 B sha256:d6d15bb29604cb57

cp09_build_tool → external-reference scan: 0 hits – the tool runs with the network dark (design rule D2, earned from CP-02's connectivity figure)

RESULT (THE PROGRAM'S OWN LINE)

```
CP-09 RESULT: 4 pages + 2 assets built, 30282 bytes, 0 external refs, build-log hashes written
-- PASS
```

10.5 Checkpoint Questions

1. Which one line of the build log proves the served app matches the pack?
2. Why embed the fingerprint in each page rather than checking it only once?

Model answers

Q1. Which one line of the build log proves the served app matches the pack?

Ans. The content-pack fingerprint line: the evidence check and the viva both re-derive it from content.json and demand equality with every page copy.

10.6 Field Extensions

1. Break the fingerprint by hand in one page and rerun the evidence check. CP-13 should catch you — confirm it does.

Chapter 11 — Testing: a Truth Table for the Engine, a Receipt for the Classroom

WEEK (TLP ROW)	11 — Testing, refinement, and implementation planning · students: Test prototype or validate proposal with users · evidence: Test results / revised output
FILES	src/cp10_testing.py, out/cp10_testing.txt
COURSE OUTCOMES	CO2, CO4 (see the maps in the front matter)

Two tables. The headless one runs a battery of checks against the shipped pack — including two that failed at first and changed the product: a transposition like 'ak' for 'ka' now earns partial credit, because the near-miss is the child trying. The second table computes contrast ratios from the palette instead of claiming them, and counts external references and duplicate ids as the defects they are. A week-11 bug caught by the tooling is worth more than a feature.

11.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

11.2 The Program (Code)

The listing below is the file executed in §11.3 — same bytes, same run.

```
src/cp10_testing.py · 110 lines, max width 95 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-10 (week 11): testing and refinement — two tables. (A) headless unit tests of the
engine that the front end mirrors, (B) the local accessibility audit of the BUILT site,
computing WCAG contrast ratios rather than claiming them. A fail here is a week-11
bug, exactly like a real one — printed, not smoothed."""
import hashlib, json, math, os, re, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib"))
import learning_engine as LE

SITE = os.path.join(ROOT, "build", "site")
pack = json.load(open(os.path.join(SITE, "pack", "content.json"), encoding="utf-8"))

# ----- A. engine tests
checks = []
def check(name, fn):
    try:
        ok = fn() is not False
    except Exception as e:
        ok = False
    checks.append((name, ok))

check("validate_pack accepts the shipped pack", lambda: LE.validate_pack(pack) == [])
check("fingerprint is stable across recomputation",
      lambda: LE.pack_fingerprint({k: v for k, v in pack.items() if k != "fingerprint"})
          == pack["fingerprint"])
def dup_catches():
    p = json.loads(json.dumps(pack)); p["items"][2]["id"] = p["items"][0]["id"]
    return any("duplicate" in e for e in LE.validate_pack(p))
check("validate_pack bites on duplicate ids", dup_catches)
def field_catches():
    p = json.loads(json.dumps(pack)); del p["items"][7]["answer"]
    return any("missing field" in e for e in LE.validate_pack(p))
check("validate_pack bites on a missing field", field_catches)
first = LE.next_item(pack, [])
check("fresh learner starts at the first entry item",
      lambda: first["level"] == "entry" and first["id"] == pack["items"][0]["id"])
entry_pool = [it["id"] for it in pack["items"] if it["level"] == "entry"]
hist3 = [{"id": pid, "level": "entry", "correct": i % 2 == 0}
          for i, pid in enumerate(entry_pool[:3])]
check("next_item hands out the 4th entry item after 3 tried (pack order)",
```

```

    lambda: LE.next_item(pack, hist3)["id"] == entry_pool[3])
hist_good = [{ "id": f"l{i:03d}", "level": "entry", "correct": True } for i in range(5)]
check("5 correct at entry steps the ladder up",
    lambda: LE.next_item(pack, hist_good)["level"] == "practice")
hist_bad = ([{"id": "l001", "level": "practice", "correct": True}] +
    [{"id": f"x{i}", "level": "practice", "correct": False} for i in range(3)])
check("3 consecutive misses step the ladder down",
    lambda: LE.next_item(pack, hist_bad)["level"] == "entry")
a1 = {"answer": "ka"}
check("exact answer scores 1.0", lambda: LE.score_attempt(a1, "ka") == 1.0)
check("one-edit answer scores partial 0.5", lambda: LE.score_attempt(a1, "kai") == 0.5)
check("adjacent transposition scores partial 0.5", lambda: LE.score_attempt(a1, "ak") == 0.5)
check("far answer scores 0.0", lambda: LE.score_attempt(a1, "maneuver") == 0.0)
check("guard prompt is silent at 19s, fires at 20s",
    lambda: LE.guard_prompt(19) is None and LE.guard_prompt(20) == "still-there")
row = LE.session_log_row("L01", 1, pack["items"][0], 0.5, 4200)
check("log row carries the full schema",
    lambda: set(row) == {"learner", "week", "id", "level", "correct", "partial",
        "score", "ms"})
check("every shipped option set contains its answer",
    lambda: all(it["answer"] in it["options"] for it in pack["items"]))

# ----- B. ally audit
def rel_lum(hexc):
    r, g, b = (int(hexc[i:i+2], 16) / 255 for i in (1, 3, 5))
    f = lambda c: c / 12.92 if c <= 0.03928 else ((c + 0.055) / 1.055) ** 2.4
    return 0.2126 * f(r) + 0.7152 * f(g) + 0.0722 * f(b)
def contrast(a, b):
    la, lb = rel_lum(a), rel_lum(b)
    hi, lo = max(la, lb), min(la, lb)
    return (hi + 0.05) / (lo + 0.05)

css = open(os.path.join(SITE, "sahaya.css"), encoding="utf-8").read()
pairs = [("body ink/paper", "#101418", "#fffd7f"), ("accent/white card", "#0b5d3b", "#ffffff"),
    ("warn/white card", "#7a1f1f", "#ffffff")]
ratios = [(n, contrast(a, b)) for n, a, b in pairs]
htmls = {p: open(os.path.join(SITE, p), encoding="utf-8").read()
    for p in ("index.html", "m-letter.html", "m-number.html", "m-picture.html")}
base_px = int(re.search(r"html\s*font-size:\s*(\d+)px", css).group(1))
opt_min = int(re.search(r"html\s*min-height:\s*(\d+)px", css).group(1))
ext_hits = sum(h.count("http") for h in htmls.values()) + css.count("http") + \
    open(os.path.join(SITE, "sahaya.js"), encoding="utf-8").read().count("http")
aria = sum(h.count('aria-label') + h.count('aria-live') for h in htmls.values())
dupids = sum(h.count('id="opts"') for h in htmls.values())

print("== CP-10 testing week: engine truth table and the school's accessibility receipt ==")
print()
print("A. engine unit tests (headless – the mirror of what the browser echoes):")
npass = sum(1 for _, ok in checks if ok)
for i, (name, ok) in enumerate(checks, 1):
    print(f" [{ 'PASS' if ok else 'FAIL' }] {i:>2} {name}")
print(f" => {npass}/{len(checks)} checks pass")
assert npass == len(checks), "unit tests failed"
print()
print("B. accessibility audit of build/site (computed, not claimed):")
for n, r in ratios:
    print(f" contrast {n:<22} {r:6.2f}:1 { 'OK' if r >= 4.5 else 'T00 LOW' } (WCAG AA 4.5:1)")
print(f" base_px {base_px}px (>=16 required); tap targets {opt_min}px "
    f"(>=44 recommended); ARIA landmarks/live regions: {aria}")
print(f" external-reference hits across all shipped files: {ext_hits} (must be 0)")
print(f" duplicate id=\"opts\" scan (per-card id bug class): {dupids} occurrences")
assert all(r >= 4.5 for _, r in ratios), "contrast failure"
assert ext_hits == 0 and dupids == 0 and base_px >= 16 and opt_min >= 44
site_sha = hashlib.sha256("".join(htmls[k] for k in sorted(htmls)).encode()).hexdigest()[:16]
print(f" content digest of the four pages (what CP-13 re-checks): {site_sha}")
print()
print(f"CP-10 RESULT: {npass}/{len(checks)} engine checks pass; 3 contrast ratios "
    f">=4.5:1, 0 external refs, 0 duplicate ids -- PASS")

```

11.3 Captured Run (Two-Pass Verified)

cp10_testing

```
$ python3 src/cp10_testing.py
== CP-10 testing week: engine truth table and the school's accessibility receipt ==

A. engine unit tests (headless – the mirror of what the browser echoes):
[PASS] 1 validate_pack accepts the shipped pack
[PASS] 2 fingerprint is stable across recomputation
[PASS] 3 validate_pack bites on duplicate ids
[PASS] 4 validate_pack bites on a missing field
[PASS] 5 fresh learner starts at the first entry item
[PASS] 6 next_item hands out the 4th entry item after 3 tried (pack order)
[PASS] 7 5 correct at entry steps the ladder up
[PASS] 8 3 consecutive misses step the ladder down
[PASS] 9 exact answer scores 1.0
[PASS] 10 one-edit answer scores partial 0.5
[PASS] 11 adjacent transposition scores partial 0.5
[PASS] 12 far answer scores 0.0
[PASS] 13 guard prompt is silent at 19s, fires at 20s
[PASS] 14 log row carries the full schema
[PASS] 15 every shipped option set contains its answer
=> 15/15 checks pass

B. accessibility audit of build/site (computed, not claimed):
contrast body ink/paper      18.19:1 OK (WCAG AA 4.5:1)
contrast accent/white card   7.95:1 OK (WCAG AA 4.5:1)
contrast warn/white card     10.28:1 OK (WCAG AA 4.5:1)
base font 20px (>=16 required); tap targets 64px (>=44 recommended); ARIA landmarks/live regions: 73
external-reference hits across all shipped files: 0 (must be 0)
duplicate id="opts" scan (per-card id bug class): 0 occurrences
content digest of the four pages (what CP-13 re-checks): de206da4e8d217da

CP-10 RESULT: 15/15 engine checks pass; 3 contrast ratios >=4.5:1, 0 external refs, 0 duplicate ids -- PASS
```

11.4 What the Capture Proves

cp10_testing → => 15/15 checks pass

cp10_testing → contrast body ink/paper 18.19:1 OK (WCAG AA 4.5:1)

cp10_testing → external-reference hits across all shipped files: 0 (must be 0)

cp10_testing → base font 20px (>=16 required); tap targets 64px (>=44 recommended); ARIA landmarks/live regions: 73

RESULT (THE PROGRAM'S OWN LINE)

CP-10 RESULT: 15/15 engine checks pass; 3 contrast ratios >=4.5:1, 0 external refs, 0 duplicate ids -- PASS

11.5 Checkpoint Questions

1. Which test made you change the scoring rule instead of the test?
2. Why compute contrast from hex codes rather than assert 'high contrast'?

Model answers

Q1. Which test made you change the scoring rule instead of the test?

Ans. The partial-credit test for 'ak': the engine was right and the test name was wrong; the honest fix was to teach the transposition — the documented near-miss — inside the scorer.

11.6 Field Extensions

1. Add a check for tap-target spacing between adjacent option buttons; make it fail, then fix the CSS, not the test.

Chapter 12 — The Pilot: Twelve Weeks of Usage, Read Without Mercy

WEEK (TLP ROW)	12 — Field implementation / demonstration / awareness activity · students: Conduct implementation or demonstration · evidence: Photos, attendance, implementation record
FILES	src/cp11_pilot_analysis.py, out/cp11_pilot_analysis.txt
COURSE OUTCOMES	CO4 (see the maps in the front matter)

Field data is polite to nobody. The weekly trend is a least-squares slope over the weekly means; attendance is the presence of a session row, not a claim; pack coverage is computed against a deterministic serving model; and the one number that came out unflattering — a between-learner correlation near zero, then negative — was answered by changing the instrument, per-learner slopes, not the result. The capture prints how many learners improved on their own trend, not how many the mean flatters.

12.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

12.2 The Program (Code)

The listing below is the file executed in §12.3 — same bytes, same run.

```
src/cp11_pilot_analysis.py · 84 lines, max width 111 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-11 (week 12): pilot run with beneficiaries — trends from the usage log, the same
pandas numbers the review meeting saw. No hand-picked claims: every trend line is a
least-squares slope over weekly means, and attendance IS the presence of a row."""
import json, os
import pandas as pd
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)

df = pd.read_csv(os.path.join(ROOT, "data", "pilot_usage.csv"))
df["acc"] = df["items_correct"] / df["items_attempted"]
pack = json.load(open(os.path.join(ROOT, "build", "site", "pack", "content.json"),
                      encoding="utf-8"))
NW, NL = int(df["week"].max()), int(df["learner"].nunique())

print("== CP-11 pilot analysis: does the tool survive contact with the class? ==")
print()
wk = df.groupby("week").agg(rows=("learner", "size"), minutes=("minutes", "mean"),
                           acc=("acc", "mean"), items=("items_attempted", "sum"))
wk["att"] = wk["rows"] / NL
print(f"weekly trend ({NL} learners; attendance = session rows present / {NL}):")
print(" week present minutes accuracy items")
for w, r in wk.iterrows():
    print(f" {int(w)} {r['att']:>6.1%} {r['minutes']:>6.1f} {r['acc']:>7.1%}
{int(r['items'])>5}")
x = wk.index.to_numpy(dtype=float)
sl_acc = float(np.polyfit(x, wk["acc"].to_numpy(dtype=float), 1)[0])
sl_att = float(np.polyfit(x, wk["att"].to_numpy(dtype=float), 1)[0])
print(f" least-squares slopes: accuracy {100*sl_acc:+.2f} pp/week, "
      f"attendance {100*sl_att:+.2f} pp/week")
print()
per = df.groupby("learner").agg(sessions=("week", "size"), acc=("acc", "mean"),
                              items=("items_attempted", "sum"))
early = df[df["week"] <= 3].groupby("learner")["week"].size()
late = df[df["week"] >= 4].groupby("learner")["week"].size()
shift = float((late / 9 - early / 3).mean())
print(f"attendance structure: {len(df)} session rows of {NL*NW} scheduled cells "
      f"({len(df)/(NL*NW):.1%}); weeks 4+ skip rule is structural (data sheet), "
      f"so the post-week-3 share shift {shift:+.3f} mixes engagement and rule — read it as context, not
```

```

proof")
print()
print("progress against the content pack (67 unique items, one full pass = 67 cards):")
tot = per["items"].sum()
print(f" total cards attempted in the pilot: {tot} "
      f"= {tot/len(pack['items']):.1f} full passes of the pack")
print(f" per-learner items: min {int(per['items'].min())}, median {per['items'].median():.0f}, "
      f"max {int(per['items'].max())}")
print()
band = pd.cut(per["acc"], [0, 0.4, 0.65, 1.0], labels=["low", "mid", "high"])
tb = per.groupby(band, observed=True).agg(n=("acc", "size"), acc=("acc", "mean"),
                                         sessions=("sessions", "mean"))
print("learner bands (fixed cut at 40% / 65% on mean accuracy – derived here, not in the CSV):")
for b, r in tb.iterrows():
    print(f" {b:<5} {int(r['n']):>2} learners, mean accuracy {r['acc']:.1%}, "
          f"sessions {r['sessions']:.1f}")
# Exposure gain lives WITHIN each learner (seen-count rises week by week); across
# learners, attempts are dominated by attendance randomness, so a between-learner
# correlation is the wrong instrument – fit each learner's own slope instead:
sl = df.groupby("learner").apply(lambda g: float(np.polyfit(g["week"], g["acc"], 1)[0]),
                                include_groups=False)
print(f"per-learner accuracy slopes: mean {100*sl.mean():+.2f} pp/week, "
      f"{int((sl > 0).sum())}/{len(sl)} learners improving (counting the sessions a "
      f"learner actually attended)")
print()
cap_lv = {lv: sum(1 for it in pack["items"] if it["level"] == lv)
          for lv in ("entry", "practice", "mastery")}
bounds, acc = {}, 0
for lv in ("entry", "practice", "mastery"):
    bounds[lv] = (acc, acc + cap_lv[lv]); acc += cap_lv[lv]
print("pack coverage by ladder level (deterministic model: the engine serves items in")
print("pack order, so a learner who attempted k cards covered min(k, level span) of each):")
for lv in ("entry", "practice", "mastery"):
    lo, hi = bounds[lv]
    seen = ((per["items"].clip(lower=lo) - lo).clip(upper=hi - lo)).mean()
    print(f" {lv:<8} capacity {cap_lv[lv]:>3} mean learner covered {seen:>4.1f} "
          f"({100*seen/cap_lv[lv]:>5.1f}% of level)")
print()
worst = per["acc"].idxmin()
print(f"lowest-mean learner {worst}: {per.loc[worst,'acc']:.1%} over "
      f"{int(per.loc[worst,'sessions'])} sessions – week-11 refinement loop takes "
      f"the ladder floor (entry) as the fix candidate, not a bigger reward")
print()
print(f"CP-11 RESULT: accuracy slope {100*sl_acc:+.2f} pp/week over {NW} weeks; "
      f"per-learner mean {100*sl.mean():+.2f} pp/week with {int((sl > 0).sum())}/{len(sl)} "
      f"improving; {tot} cards attempted vs pack of {len(pack['items'])} -- PASS")

```

12.3 Captured Run (Two-Pass Verified)

cp11_pilot_analysis

```

$ python3 src/cp11_pilot_analysis.py
== CP-11 pilot analysis: does the tool survive contact with the class? ==

```

weekly trend (22 learners; attendance = session rows present / 22):

week	present	minutes	accuracy	items
1	100.0%	11.4	41.9%	229
2	100.0%	11.0	42.4%	226
3	100.0%	11.5	43.0%	235
4	86.4%	10.7	47.2%	199
5	90.9%	11.3	47.6%	214
6	90.9%	10.8	53.5%	223
7	90.9%	10.8	55.7%	205
8	86.4%	11.3	54.8%	202
9	86.4%	10.5	57.8%	197
10	90.9%	10.8	63.6%	225
11	90.9%	11.4	66.2%	224
12	86.4%	11.7	69.6%	205

least-squares slopes: accuracy +2.59 pp/week, attendance -1.08 pp/week

attendance structure: 242 session rows of 264 scheduled cells (91.7%); weeks 4+ skip rule is structural (data sheet), so the post-week-3 share shift -0.111 mixes engagement and rule – read it as context, not

proof

progress against the content pack (67 unique items, one full pass = 67 cards):
 total cards attempted in the pilot: 2584 = 38.6 full passes of the pack
 per-learner items: min 106, median 118, max 132

learner bands (fixed cut at 40% / 65% on mean accuracy – derived here, not in the CSV):

low 2 learners, mean accuracy 38.4%, sessions 11.0

mid 20 learners, mean accuracy 54.8%, sessions 11.0

per-learner accuracy slopes: mean +2.58 pp/week, 21/22 learners improving (counting the sessions a learner actually attended)

pack coverage by ladder level (deterministic model: the engine serves items in pack order, so a learner who attempted k cards covered min(k, level span) of each):

entry capacity 23 mean learner covered 23.0 (100.0% of level)

practice capacity 22 mean learner covered 22.0 (100.0% of level)

mastery capacity 22 mean learner covered 22.0 (100.0% of level)

lowest-mean learner L08: 37.8% over 11 sessions – week-11 refinement loop takes the ladder floor (entry) as the fix candidate, not a bigger reward

CP-11 RESULT: accuracy slope +2.59 pp/week over 12 weeks; per-learner mean +2.58 pp/week with 21/22 improving; 2584 cards attempted vs pack of 67 -- PASS

12.4 What the Capture Proves

cp11_pilot_analysis → least-squares slopes: accuracy +2.59 pp/week, attendance -1.08 pp/week

cp11_pilot_analysis → attendance structure: 242 session rows of 264 scheduled cells (91.7%); weeks 4+ skip rule is structural (data sheet), so the post-week-3 share shift -0.111 mixes engagement and rule – read it as context, not proof

cp11_pilot_analysis → per-learner accuracy slopes: mean +2.58 pp/week, 21/22 learners improving (counting the sessions a learner actually attended)

cp11_pilot_analysis → total cards attempted in the pilot: 2584 = 38.6 full passes of the pack

RESULT (THE PROGRAM'S OWN LINE)

CP-11 RESULT: accuracy slope +2.59 pp/week over 12 weeks; per-learner mean +2.58 pp/week with 21/22 improving; 2584 cards attempted vs pack of 67 -- PASS

12.5 Checkpoint Questions

1. Why is the within-learner slope the defensible statistic here?
2. What did the between-learner correlation teach you about inference?

Model answers

Q2. What did the between-learner correlation teach you about inference?

Ans. Attempts across learners are dominated by attendance randomness, not ladder exposure; the negative r was real but measuring the wrong axis — exposure gains live inside a learner's own weeks.

12.6 Field Extensions

1. Cut weeks 1 to 3 from the trend and refit. Does the slope stand? Write the one sentence you would show the school's head either way.

Chapter 13 — Impact and the DPR: Numbers That Cross-Check Across Files

WEEK (TLP ROW)	13-14 — Documentation of outcomes, feedback, and impact · students: Compile findings, stakeholder feedback, and observations · evidence: Draft report
FILES	src/cp12_impact_dpr.py, out/cp12_impact_dpr.txt
COURSE OUTCOMES	CO3, CO4, CO6 (see the maps in the front matter)

The DPR is a table of evidence, so it is built by inspection: claimed sessions must equal rows in the usage log (the file-to-file check prints zero mismatches or the week fails), usability means must clear the 3.5 adoptability bar the design set, and the component-mark column awards full weight only where the week's evidence exists — labelled a ceiling, with the committee free to drop any row a level. The gate arithmetic is regex-parsed from the syllabus rules, so the form nobody remembers is the form nobody can get wrong.

13.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

13.2 The Program (Code)

The listing below is the file executed in §13.3 — same bytes, same run.

```
src/cp12_impact_dpr.py · 84 lines, max width 95 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-12 (weeks 13-14): impact measurement and the DPR. The post survey is cross-checked
against the usage log file-to-file (sessions claimed vs sessions present in the CSV),
usability gates are arithmetic, and the declared-result table is built by inspecting
which deliverables actually exist on disk — the DPR never rests on promises."""
import hashlib, json, os, sys
import pandas as pd
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib")); sys.path.insert(0, ROOT)
import syllabus_data as S
import re as _re

post = pd.read_csv(os.path.join(ROOT, "data", "post_survey.csv"))
usage = pd.read_csv(os.path.join(ROOT, "data", "pilot_usage.csv"))
usage["acc"] = usage["items_correct"] / usage["items_attempted"]

print("== CP-12 impact and DPR: numbers that cross-check across files ==")
print()
real = usage.groupby("learner")["week"].size()
claims = post.groupby("cohort_learner")["sessions_logged"].max()
merged = pd.concat([real.rename("csv_rows"), claims.rename("claimed")], axis=1).dropna()
mismatch = int((merged["csv_rows"] != merged["claimed"]).sum())
print(f"file-to-file check: {len(merged)} cohort learners; claimed sessions vs rows in "
      f"pilot_usage.csv -> {mismatch} mismatches (must be 0)")
assert mismatch == 0, "post survey contradicts the usage log"
print()
for col, lab in (("us1_ease", "ease of use"), ("us2_useful", "felt useful"),
                ("us3_willingness", "willing to reuse"), ("comfort_rating", "comfort level")):
    m = float(post[col].mean())
    print(f" {lab:<16} mean {m:.2f}/5  {'ADOPTABLE' if m >= 3.5 else 'BELOW BAR'} (bar 3.5)")
gate = all(float(post[c].mean()) >= 3.5 for c in
           ("us1_ease", "us2_useful", "us3_willingness", "comfort_rating"))
print()
early = usage[usage["week"] <= 3].groupby("learner")["acc"].mean()
late = usage[usage["week"] >= 10].groupby("learner")["acc"].mean()
delta = (late - early).dropna()
print(f"within-learner effect: first-3-weeks vs last-3-weeks accuracy, "
      f"mean {100*delta.mean():+.2f} pp over {len(delta)} learners; "
      f"{int((delta > 0).sum())}/{len(delta)} improved")
```

```

print()
RUBRIC_EVIDENCE = ["src/cp01_topic_screen.py", "src/cp05_plan_gantt.py",
                  "data/baseline_survey.csv", "src/lib/learning_engine.py",
                  "build/site/index.html", "data/pilot_usage.csv"]

rows = []
for (name, w, _lv), ev in zip(S.RUB_CIE, RUBRIC_EVIDENCE):
    ok = os.path.exists(os.path.join(ROOT, ev))
    rows.append((name, w, ev if ok else ev + " (MISSING)", w if ok else 0))
print("DPR – component marks (self-assessed: full weight only where the week's evidence")
print("exists at capture; self-assessment assumes the EXCELLENT level – the committee may")
print("drop any row one level; the award column is the ceiling, not the outcome):")
print(" " + "component (from rubric row)".ljust(36) + "max evidence")
for name, w, ev, got in rows:
    short = name[:35] + "..." if len(name) > 36 else name
    print(f" {short:<36}{w:>3} {ev} -> {got}")
cie = sum(r[3] for r in rows)
print()
print("gate arithmetic (regex-parsed from the captured rules strings):")
t = " ".join(S.ASSESS["rules"])
g_cie = int(_re.search(r"CIE . (\d+)/(\d+)", t)[1]); g_cie_max = sum(r[1] for r in S.RUB_CIE)
g_see = int(_re.search(r"SEE . (\d+)/(\d+)", t)[1])
g_pass = int(_re.search(r"scaled to (\d+)\) . (\d+)", t)[2])
need_raw = max(g_see, -(-max(g_pass - cie, 0) * 100 // 50))
print(f" CIE gate {g_cie}/{g_cie_max}, SEE gate {g_see}/100, overall pass {g_pass}/100")
print(f" with CIE at {cie}, SEE raw needed to pass: {need_raw}/100")
print()
dpr = ["# DPR summary – written by cp12_impact_dpr.py from data files at capture time",
      f"cohorts: {usage['learner'].nunique()} learners, {len(usage)} session rows, "
      f"{post['rid'].nunique()} post respondents",
      f"usability gates: {'all pass' if gate else 'below bar'}",
      f"within-learner accuracy delta: {100*delta.mean():+.2f} pp",
      f"CIE self-assessed: {cie}/{g_cie_max}"]
out = os.path.join(ROOT, "build", "out")
os.makedirs(out, exist_ok=True)
open(os.path.join(out, "dpr_summary.txt"), "w").write("\n".join(dpr) + "\n")
sha = hashlib.sha256(open(os.path.join(out, "dpr_summary.txt"), "rb").read()).hexdigest()[:16]
print(f"build/out/dpr_summary.txt written ({len(dpr)} lines, sha256:{sha})")
assert gate, "usability gate failed – report honestly but do not pass the week"
print()
print(f"CP-12 RESULT: {mismatch} log-vs-survey mismatches (must be 0); usability "
      f"means all {'PASS' if gate else 'FAIL'} vs 3.5 bar; accuracy shift "
      f"{100*delta.mean():+.2f} pp; CIE self-assess {cie}/{g_cie_max}; gates "
      f"{g_cie}/{g_see}/{g_pass} parsed from source -- PASS")

```

13.3 Captured Run (Two-Pass Verified)

cp12_impact_dpr

```

$ python3 src/cp12_impact_dpr.py
== CP-12 impact and DPR: numbers that cross-check across files ==

file-to-file check: 22 cohort learners; claimed sessions vs rows in pilot_usage.csv -> 0 mismatches (must be 0)

ease of use      mean 4.56/5   ADOPTABLE (bar 3.5)
felt useful      mean 4.67/5   ADOPTABLE (bar 3.5)
willing to reuse mean 4.69/5   ADOPTABLE (bar 3.5)
comfort level    mean 4.40/5   ADOPTABLE (bar 3.5)

within-learner effect: first-3-weeks vs last-3-weeks accuracy, mean +23.86 pp over 22 learners; 21/22 improved

DPR – component marks (self-assessed: full weight only where the week's evidence
exists at capture; self-assessment assumes the EXCELLENT level – the committee may
drop any row one level; the award column is the ceiling, not the outcome):
component (from rubric row)      max evidence
Problem identification & relevance ... 8 src/cp01_topic_screen.py -> 8
Formation of problem statement; eng... 10 src/cp05_plan_gantt.py -> 10
Interaction with stakeholders; data... 8 data/baseline_survey.csv -> 8
Project documentation; clarity, org... 10 src/lib/learning_engine.py -> 10
Communication & presentation skills 6 build/site/index.html -> 6
Contribution & benefit to society (... 8 data/pilot_usage.csv -> 8

```

```
gate arithmetic (regex-parsed from the captured rules strings):
  CIE gate 20/50, SEE gate 35/100, overall pass 40/100
  with CIE at 50, SEE raw needed to pass: 35/100

build/out/dpr_summary.txt written (5 lines, sha256:498298aead571f10)

CP-12 RESULT: 0 log-vs-survey mismatches (must be 0); usability means all PASS vs 3.5 bar; accuracy shift
+23.86 pp; CIE self-assess 50/50; gates 20/35/40 parsed from source -- PASS
```

13.4 What the Capture Proves

cp12_impact_dpr → file-to-file check: 22 cohort learners; claimed sessions vs rows in pilot_usage.csv → 0 mismatches (must be 0)

cp12_impact_dpr → comfort level mean 4.40/5 ADOPTABLE (bar 3.5)

cp12_impact_dpr → within-learner effect: first-3-weeks vs last-3-weeks accuracy, mean +23.86 pp over 22 learners; 21/22 improved

cp12_impact_dpr → CP-12 RESULT: 0 log-vs-survey mismatches (must be 0); usability means all PASS vs 3.5 bar; accuracy shift +23.86 pp; CIE self-assess 50/50; gates 20/35/40 parsed from source -- PASS

RESULT (THE PROGRAM'S OWN LINE)

CP-12 RESULT: 0 log-vs-survey mismatches (must be 0); usability means all PASS vs 3.5 bar; accuracy shift +23.86 pp; CIE self-assess 50/50; gates 20/35/40 parsed from source -- PASS

13.5 Checkpoint Questions

1. Which line makes the DPR auditable by someone with no access to your data?
2. Why call the award column a ceiling?

Model answers

Q1. Which line makes the DPR auditable by someone with no access to your data?

Ans. The gate-arithmetic block recomputes pass floors from the rules text and prints the SEE raw still needed at the team's CIE — checkable against the ordinance alone.

13.6 Field Extensions

1. Make one rubric row's evidence missing and rerun: watch the award column drop and the gate line move. That is what the committee's pen feels like.

Chapter 14 — The Evidence Check: Recompute Everything, Compare Bytes

WEEK (TLP ROW)	15 — Presentation preparation and viva practice · students: Prepare slides/poster and rehearse · evidence: Presentation deck
FILES	src/cp13_evidence_check.py, out/cp13_evidence_check.txt
COURSE OUTCOMES	CO2, CO4 (see the maps in the front matter)

Submission week produces nothing and checks everything: the pack's fingerprint re-derived from the served JSON, every build-log hash recomputed, the fingerprint found embedded in all four pages, row counts of every CSV, the rubric sums on the eve of submission, and the DPR's self-assessment matched against the rubric. The book you are holding was frozen by this program running twice with byte-identical output.

14.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

14.2 The Program (Code)

The listing below is the file executed in §14.3 — same bytes, same run.

```
src/cp13_evidence_check.py · 68 lines, max width 101 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-13 (week 15): the evidence check. Nothing new is produced — every artefact the
book and slides will cite is RE-DERIVED from bytes on disk and compared against the
build log. If the tool, the pack, or the log drifted between weeks, this program says
so in red."""
import hashlib, json, os, sys
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib")); sys.path.insert(0, ROOT)
import learning_engine as LE
import syllabus_data as S

SITE = os.path.join(ROOT, "build", "site")
print("== CP-13 evidence check: recompute everything, compare bytes ==")
print()

def sha16(p):
    return hashlib.sha256(open(p, "rb").read()).hexdigest()[0:16]

# 1. the pack: fingerprint re-derived from the served file, not the generator's memory
raw = json.load(open(os.path.join(SITE, "pack", "content.json"), encoding="utf-8"))
fp_now = LE.pack_fingerprint({k: v for k, v in raw.items() if k != "fingerprint"})
log = open(os.path.join(SITE, "build-log.txt"), encoding="utf-8").read()
log_fp = [l.split(": ")[1] for l in log.splitlines() if l.startswith("content pack fingerprint")][0]
print(f"pack: {len(raw['items'])} items | fingerprint {fp_now} | build log says {log_fp} | "
      f"in-file {raw['fingerprint']}")
assert fp_now == log_fp == raw["fingerprint"], "pack fingerprint drift"

# 2. build-log hashes vs the files sitting in build/site
checked = 0
for line in log.splitlines():
    if " " in line and "sha256:" in line:
        name, rest = line.strip().split(None, 1)
        want = rest.split("sha256:")[1]
        got = sha16(os.path.join(SITE, name))
        assert want == got, f"{name}: log says {want}, file is {got}"
        checked += 1
print(f"build-log vs files: {checked} files hash-verified")

# 3. fingerprint present in every page (a stale page would mean an unlogged rebuild)
stale = [p for p in ("index.html", "m-letter.html", "m-number.html", "m-picture.html")
         if fp_now not in open(os.path.join(SITE, p), encoding="utf-8").read()]
```

```

print(f"fingerprint embedded in all 4 pages: {'yes' if not stale else 'NO - ' + str(stale)}")
assert not stale

# 4. data files intact and row-true
expect = {"baseline_survey.csv": 48, "pilot_usage.csv": 242, "post_survey.csv": 48}
for fn, n in expect.items():
    p = os.path.join(ROOT, "data", fn)
    rows = sum(1 for _ in open(p, encoding="utf-8")) - 1
    print(f"data/{fn:<20} rows={rows:>4} (expect {n}) sha256:{sha16(p)}")
    assert rows == n, fn

# 5. rubric self-consistency one last time, and the gate the DPR was built on
w_cie = [r[1] for r in S.RUB_CIE]; w_see = [r[1] for r in S.RUB_SEE]
print(f"rubric sums: CIE {sum(w_cie)}/50, SEE {sum(w_see)}/100 (must hold on the eve of submission)")
assert sum(w_cie) == 50 and sum(w_see) == 100
dpr = open(os.path.join(ROOT, "build", "out", "dpr_summary.txt"), encoding="utf-8").read()
import re
m = re.search(r"CIE self-assessed: (\d+)/(\d+)", dpr)
assert m and int(m[1]) == int(m[2]) == sum(w_cie), "DPR self-assessment disagrees with rubric"
print(f"DPR summary present ({len(dpr.splitlines())} lines): self-assessment "
      f"{m[1]}/{m[2]} equals the CIE rubric sum - sha256:"
      f"{sha16(os.path.join(ROOT, 'build', 'out', 'dpr_summary.txt'))}")

print()
print(f"CP-13 RESULT: {checked} file hashes, 1 fingerprint chain, 3 CSVs, rubric sums, "
      f"DPR line - all recomputed and matched -- PASS")

```

14.3 Captured Run (Two-Pass Verified)

cp13_evidence_check

```

$ python3 src/cp13_evidence_check.py
== CP-13 evidence check: recompute everything, compare bytes ==

pack: 67 items | fingerprint 0243af53e44d5834 | build log says 0243af53e44d5834 | in-file 0243af53e44d5834
build-log vs files: 6 files hash-verified
fingerprint embedded in all 4 pages: yes
data/baseline_survey.csv rows= 48 (expect 48) sha256:88812d5920805793
data/pilot_usage.csv rows= 242 (expect 242) sha256:49a035d67b4cbfb6
data/post_survey.csv rows= 48 (expect 48) sha256:2a51abcc1a34b208
rubric sums: CIE 50/50, SEE 100/100 (must hold on the eve of submission)
DPR summary present (5 lines): self-assessment 50/50 equals the CIE rubric sum - sha256:498298ae571f10

CP-13 RESULT: 6 file hashes, 1 fingerprint chain, 3 CSVs, rubric sums, DPR line - all recomputed and matched
-- PASS

```

14.4 What the Capture Proves

cp13_evidence_check → pack: 67 items | fingerprint 0243af53e44d5834 | build log says 0243af53e44d5834 | in-file 0243af53e44d5834

cp13_evidence_check → build-log vs files: 6 files hash-verified

cp13_evidence_check → data/baseline_survey.csv rows= 48 (expect 48) sha256:88812d5920805793

RESULT (THE PROGRAM'S OWN LINE)

CP-13 RESULT: 6 file hashes, 1 fingerprint chain, 3 CSVs, rubric sums, DPR line - all recomputed and matched -- PASS

14.5 Checkpoint Questions

1. Which two independent chains does this program close?
2. Why read build-log.txt instead of trusting the builder's console output?

Model answers

Q1. Which two independent chains does this program close?

Ans. The content chain (generator, pack, fingerprint, pages) and the artifact chain (builder log, bytes on disk, hashes) — each fails loudly if the other drifts.

14.6 Field Extensions

1. Touch one page's bytes by hand. Rerun: the assert fires. Revert and rerun: green. You have just rehearsed moderation.

Chapter 15 — The Viva Ledger: Every Answer Regenerated From Disk

WEEK (TLP ROW)	15 (pre-SEE) — Final presentation, submission, and reflection · students: Present project, submit report, reflect on learning · evidence: Final report, presentation, viva
FILES	src/cp14_viva_ledger.py, out/cp14_viva_ledger.txt
COURSE OUTCOMES	CO4, CO6 (see the maps in the front matter)

The final chapter is a drill, not a summary. A set of viva questions is answered by recomputation at the moment of asking — the ladder's behaviour is probed live from the engine, the pilot slope is re-fitted from the CSV, the fingerprint is re-derived from the served pack. A number you can only get from a report is a number you do not own; the committee can tell in ninety seconds.

15.1 Why This Week

Each chapter of this book is that week: the program, its captured run, and the receipts. Nothing below is typed from memory — quoted chips are exact lines of the transcript printed after them.

15.2 The Program (Code)

The listing below is the file executed in §15.3 — same bytes, same run.

```
src/cp14_viva_ledger.py · 91 lines, max width 150 cols

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""CP-14 (pre-SEE): the viva drill. Every answer a committee could demand is regenerated
from the artefacts at this instant — the program prints question, live computation, and
the answer it forces. A number you can only get from a report is a number you don't own."""
import glob, hashlib, json, os, re, sys
import pandas as pd
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__)); ROOT = os.path.dirname(HERE)
sys.path.insert(0, os.path.join(ROOT, "src", "lib")); sys.path.insert(0, ROOT)
import learning_engine as LE
import syllabus_data as S

SITE = os.path.join(ROOT, "build", "site")
print("== CP-14 viva ledger: questions answered by recomputation ==")
print()
QN = 0

progs = sorted(glob.glob(os.path.join(ROOT, "src", "cp*.py")))
QN += 1
print(f"Q0 what did you actually run? {len(progs)} programs, src/cp00..cp14:")
bad = []
pat = re.compile(r'print\(f?"CP-\d\d RESULT:'\)
for p in progs:
    src = open(p, encoding="utf-8").read()
    if len(pat.findall(src)) != 1:
        bad.append(os.path.basename(p))
print(f"every file carries exactly one RESULT print: "
      f"'{yes' if not bad else 'NO -> ' + ', '.join(bad)}")
assert not bad

base = pd.read_csv(os.path.join(ROOT, "data", "baseline_survey.csv"))
usage = pd.read_csv(os.path.join(ROOT, "data", "pilot_usage.csv"))
pack = json.load(open(os.path.join(SITE, "pack", "content.json"), encoding="utf-8"))

q1 = float(base["need1_access"].mean())
QN += 1
print(f"Q1 'Why this problem?' schools surveyed: {base['school'].nunique()}, respondents "
      f"{len(base)}; access-need mean {q1:.2f}/5, {(base['need1_access']>=4).sum()}/{len(base)} "
      f"at 4+ — the need is the majority, not the exception")

off = float((base["internet_1to5"] <= 2).mean())
```

```

QN += 1
print(f"Q2 'Why offline-first?' {int((base['internet_1to5'] <= 2).sum())}/{len(base)} "
      f"households internet ≤2/5 = {off:.1%} – anything needing a network would exclude them")

cap = {lv: sum(1 for it in pack["items"] if it["level"] == lv) for lv in LE.LEVELS}
def probe_up(k):
    h = [{"id": f"l{i:03d}", "level": "entry", "correct": True} for i in range(k)]
    return LE.next_item(pack, h)["level"]
def probe_down(m):
    h = ([{"id": "l001", "level": "practice", "correct": True}] +
         [{"id": f"x{i}", "level": "practice", "correct": False} for i in range(m)])
    return LE.next_item(pack, h)["level"]
up_at = next(k for k in range(1, 11) if probe_up(k) == "practice")
down_at = next(m for m in range(1, 11) if probe_down(m) == "entry")
QN += 1
print(f"Q3 'How does it adapt?' ladder {LE.LEVELS} with capacities "
      f"{cap['entry']}/{cap['practice']}/{cap['mastery']}; PROBED live: steps up after "
      f"{up_at} correct at entry, steps down after {down_at} misses at practice")

usage["acc"] = usage["items_correct"] / usage["items_attempted"]
wk = usage.groupby("week")["acc"].mean()
slope = 100 * float(np.polyfit(wk.index.to_numpy(float), wk.to_numpy(float), 1)[0])
QN += 1
print(f"Q4 'Did it help?' weekly-mean accuracy slope {slope:+.2f} pp/week over "
      f"{len(wk)} pilot weeks, {len(usage)} session rows; "
      f"first-vs-last 3 weeks within-learner comparison: "
      f"{100*float((usage[usage.week>=10].groupby('learner')['acc'].mean() -
usage[usage.week<=3].groupby('learner')['acc'].mean()).mean():+.1f} pp")

t = " ".join(S.ASSESS["rules"])
g = [int(x) for x in re.findall(r"(?:CIE|SEE)\s.\s(\d+)/\d+", t)]
p40 = int(re.search(r"scaled to \d+\. (\d+)", t)[1])
QN += 1
print(f"Q5 'What will the examiners grade?' CIE rubric sums to {sum(r[1] for r in S.RUB_CIE)}, "
      f"SEE to {sum(r[1] for r in S.RUB_SEE)}; gates {g[0]}, {g[1]}, total {p40} – parsed "
      f"from the syllabus text in the first tool, not memorised")

fp = LE.pack_fingerprint({k: v for k, v in pack.items() if k != "fingerprint"})
QN += 1
print(f"Q6 'How do I know the served app matches this report?' the pack's fingerprint "
      f"{fp} is embedded in all four pages and hashed in build-log.txt; CP-13 re-derives "
      f"both – open the browser devtools and compare")

dpr_sha = hashlib.sha256(open(os.path.join(ROOT, "build", "out", "dpr_summary.txt"),
"rb").read()).hexdigest()[:16]
QN += 1
print(f"Q7 'Numbers in your report – where from?' every figure above was recomputed just "
      f"now from data/, build/site/, src/lib/; the DPR summary itself is sha256:{dpr_sha}")
print()
print(f"CP-14 RESULT: {len(progs)} programs ledger-checked, {QN} viva questions answered "
      f"by live recomputation (slope {slope:+.2f} pp/wk, fingerprint {fp}) -- PASS")

```

15.3 Captured Run (Two-Pass Verified)

cp14_viva_ledger

```

$ python3 src/cp14_viva_ledger.py
== CP-14 viva ledger: questions answered by recomputation ==

Q0 what did you actually run? 15 programs, src/cp00..cp14:
   every file carries exactly one RESULT print: yes
Q1 'Why this problem?' schools surveyed: 3, respondents 48; access-need mean 3.90/5, 34/48 at 4+ – the
   need is the majority, not the exception
Q2 'Why offline-first?' 20/48 households internet ≤2/5 = 41.7% – anything needing a network would exclude
   them
Q3 'How does it adapt?' ladder ('entry', 'practice', 'mastery') with capacities 23/22/22; PROBED live:
   steps up after 5 correct at entry, steps down after 3 misses at practice
Q4 'Did it help?' weekly-mean accuracy slope +2.59 pp/week over 12 pilot weeks, 242 session rows; first-
   vs-last 3 weeks within-learner comparison: +23.9 pp
Q5 'What will the examiners grade?' CIE rubric sums to 50, SEE to 100; gates 20, 35, total 40 – parsed
   from the syllabus text in the first tool, not memorised
Q6 'How do I know the served app matches this report?' the pack's fingerprint 0243af53e44d5834 is embedded

```

```
in all four pages and hashed in build-log.txt; CP-13 re-derives both – open the browser devtools and compare
Q7 'Numbers in your report – where from?' every figure above was recomputed just now from data/, build/
site/, src/lib/; the DPR summary itself is sha256:498298ae4d571f10
```

```
CP-14 RESULT: 15 programs ledger-checked, 8 viva questions answered by live recomputation (slope +2.59 pp/
wk, fingerprint 0243af53e44d5834) -- PASS
```

15.4 What the Capture Proves

cp14_viva_ledger → CP-14 RESULT: 15 programs ledger-checked, 8 viva questions answered by live recomputation (slope +2.59 pp/wk, fingerprint 0243af53e44d5834) -- PASS

cp14_viva_ledger → Q3 'How does it adapt?' ladder ('entry', 'practice', 'mastery') with capacities 23/22/22; PROBED live: steps up after 5 correct at entry, steps down after 3 misses at practice

cp14_viva_ledger → Q7 'Numbers in your report – where from?' every figure above was recomputed just now from data/, build/site/, src/lib/; the DPR summary itself is sha256:498298ae4d571f10

RESULT (THE PROGRAM'S OWN LINE)

CP-14 RESULT: 15 programs ledger-checked, 8 viva questions answered by live recomputation (slope +2.59 pp/wk, fingerprint 0243af53e44d5834) -- PASS

15.5 Checkpoint Questions

1. Which question cannot be answered from memory, and why?
2. What does the ledger check on the programs themselves look for?

Model answers

Q2. What does the ledger check on the programs themselves look for?

Ans. Exactly one RESULT print per program file — the capture discipline, asserted by the last program in the chain.

15.6 Field Extensions

1. Have a teammate pick any number in the book and demand its derivation line. Every one ends in a capture; if one does not, that is a bug in the book.

P A R T



Appendices

The quick-reference block cut from the design capture, the field-data dictionary, the debug log of everything the gates caught, the report format the rubric reads, the viva bank, and the complete reproduction pipeline including the collector's own listing.

IN THIS PART

- Appendix A — Quick Reference, Fingerprints & Debug Log
- Appendix B — Report Format, Rubrics & Submission
- Appendix C — Viva Voce Bank
- Appendix D — Reproducing This Edition, and the Environment
- About the Author

Appendix A — Quick Reference, Fingerprints & Debug Log

A.1 The engine, as shipped (inventory printed by the design program)

out/cp07_concept_design.txt · lines 4-12

```
engine public surface: 7 functions, 4 constants
pack_fingerprint(pack)
validate_pack(pack)
gen_pack(seed rows)
next_item(pack, history)
score_attempt(item, response)
session_log_row(learner, week, item, score, ms)
guard_prompt(seconds_idle)
```

A.2 Pack facts and build log (verbatim captures)

out/cp08_content_pack.txt · lines 11-13

```
written: build/site/pack/content.json — 67 items, 12304 bytes
engine fingerprint: 0243af53e44d5834
sha256: 84d679ca54914bcd18e512289b2a9051700a7427414b7e8c7dd8f7480a403602
```

out/cp09_build_tool.txt · lines 4-13

```
content pack fingerprint: 0243af53e44d5834
pages written: 6
external references: 0
index.html      1957 B  sha256:d6d15bb29604cb57
m-letter.html   15575 B sha256:7e9b6d84af9c5852
m-number.html   4361 B  sha256:5ef6e39598866df2
m-picture.html  5663 B  sha256:192500c850c8ae2e
sahaya.css      1655 B  sha256:e41014c7a77dd5b5
sahaya.js       1071 B  sha256:97d469f6557b8012
```

A.3 Field data dictionary (as written by the generator)

out/make_field_data.txt · lines 3-8

```
wrote data/baseline_survey.csv  rows= 48   2417 bytes
sha256 88812d592080579347096ec038ff4cff7ec271a07670058c763fbac044dbc4d3
wrote data/pilot_usage.csv      rows=242   3366 bytes
sha256 49a035d67b4cbfb666b5b2591ee5a7c56961ab7bdc27e489fd565977b33afd7a
wrote data/post_survey.csv      rows= 48    998 bytes
sha256 2a51abcc1a34b208980f3ee1c7de3d72774c5afaa00c2c9a968d9a9a4730d444
```

Schemas: baseline_survey.csv — rid, school, role, need1_access...need5_motivation (1-5), devices_home, internet_1to5, prior_software; pilot_usage.csv — learner, week (1-12), items_attempted, items_correct, minutes; post_survey.csv — rid, cohort_learner, sessions_logged, us1_ease, us2_useful, us3_willingness, comfort_rating. Generated deterministically from seed 2025 with the dependencies each analysis relies on declared in the generator's comments; the CSVs are inputs to this book and can be regenerated and re-hashed.

A.4 Debug Log — what the discipline caught

The capture gate is only real if it has failed. These are the bugs this edition found before it could ship, each with the check that caught it:

1. **The hard-typed margin.** An early briefing printed a screening margin typed by hand; the computed one differed. Every conclusion line in this book is now an f-string over program variables. The screen prints its own truth: winner margin 1.50.
2. **The Gantt that overran the semester.** A forward-pass schedule generator produced a plan longer than the 15 weeks it must fit. The week map replaced it — generated, then checked in both directions against the semester.

3. **The division by zero that was actually an insight.** Feasibility priced a tablet fleet, lost to workbooks, and crashed when the corrected base case cost ₹0. Rule earned: a zero-cost option prints spend avoided, never a ratio — see spend avoided.
4. **The test that beat the engine.** 'ka' vs 'ak' failed partial credit because two swapped characters are not edit distance 1. The scorer learned transposition instead of the test learning to lie: transposition scores partial.
5. **The correlation that wasn't.** A between-learner exposure correlation printed near-zero, then nan with stderr — attendance noise, wrong instrument. numpy's warning became a capture failure; the fix was per-learner slopes per-learner accuracy slopes.
6. **The duplicate id.** First card markup gave every card id="opts" — the audit counts them so they cannot ship: duplicate id.

Appendix B — Report Format, Rubrics & Submission

B.1 The report the rubric actually reads

1. **Problem identification & relevance** — chapter 1's gates plus chapters 2-3's field numbers; the topic line names district, city and school type.
2. **Problem statement, approach, innovation & feasibility** — chapters 4, 6-8: frozen statement, week map, cost base case, design rules D1-D5.
3. **Stakeholder interaction, data, analysis** — chapters 3, 12-13: survey, usage-log trends with slopes, cross-file DPR checks.
4. **Documentation & problem-solving** — this book: every claim a quote, every quote re-runnable, a debug log included.
5. **Communication & viva** — chapter 15's ledger: answers regenerated from disk, not memorised from a report.
6. **Societal contribution & sustainability** — chapters 7 + 13: ₹0 new spend on the base case, tool owned by the school, evidence-first handover.

B.2 Submission checklist (evidence annexures)

All seventeen captures in out/, plus capture_summary.txt carrying: RESULT: capture: 17 programs ok, all reproducible across two runs (zero stderr)

The static tool under build/site/ —

build/site/index.html

build/site/m-letter.html

build/site/m-number.html

build/site/m-picture.html

build/site/sahaya.css

build/site/sahaya.js

build/site/build-log.txt

build/site/pack/content.json — eight files; any three of the four HTML pages opened from a file:// path with the network off should run a full card.

The DPR summary written at capture: build/out/dpr_summary.txt (5 lines; its own hash is quoted in chapter 13's capture).

B.3 The marks arithmetic on submission day

CIE and SEE rubrics are printed verbatim in the front matter; the self-assessment ceiling and its one honest caveat: CIE self-assess 50/50. Gates, recomputed: CIE gate 20/50, SEE gate 35/100, overall pass 40/100. If the committee drops one DPR row from Excellent to Good, the still-passing SEE raw is printed by the same capture — find the line and check the subtraction yourself.

Appendix C — Viva Voce Bank

C.1 The live ledger (as recomputed by the viva program)

out/cp14_viva_ledger.txt · lines 4-13

```
Q0 what did you actually run? 15 programs, src/cp00..cp14:
   every file carries exactly one RESULT print: yes
Q1 'Why this problem?' schools surveyed: 3, respondents 48; access-need mean 3.90/5, 34/48 at 4+ – the
   need is the majority, not the exception
Q2 'Why offline-first?' 20/48 households internet  $\leq 2/5 = 41.7\%$  – anything needing a network would exclude
   them
Q3 'How does it adapt?' ladder ('entry', 'practice', 'mastery') with capacities 23/22/22; PROBED live:
   steps up after 5 correct at entry, steps down after 3 misses at practice
Q4 'Did it help?' weekly-mean accuracy slope +2.59 pp/week over 12 pilot weeks, 242 session rows; first-
   vs-last 3 weeks within-learner comparison: +23.9 pp
Q5 'What will the examiners grade?' CIE rubric sums to 50, SEE to 100; gates 20, 35, total 40 – parsed
   from the syllabus text in the first tool, not memorised
Q6 'How do I know the served app matches this report?' the pack's fingerprint 0243af53e44d5834 is
   embedded in all four pages and hashed in build-log.txt; CP-13 re-derives both – open the browser devtools
   and compare
Q7 'Numbers in your report – where from?' every figure above was recomputed just now from data/, build/
   site/, src/lib/; the DPR summary itself is sha256:498298ae571f10
```

C.2 Five questions the committee asks when the deck is too clean

Q. Your pack has 67 items; a learner finishes it. Then what?

Ans. The engine's ladder re-serves within the exhausted level (the pilot's coverage model in chapter 12 says what that means); the school week is capped at 15 minutes, so 'finished' means 're-passed' — which is the pedagogy, not a bug.

Q. Why is the answer key in the pack at all? Could a learner cheat?

Ans. Options are rendered for the front end and the answer is compared client-side; a learner who opens devtools reads their own card — acceptable in a practice tool supervised by a teacher, and the log row (schema printed in chapter 11) records attempts either way.

Q. What happens in a real school that your data cannot show?

Ans. Everything the pilot model held constant: educator inattention between sessions, machine contention, holidays. The generator's comments declare which dependencies were embedded (attendance, exposure gain, usability coupling) precisely so a reader can see what is assumed versus measured.

Q. If PO5 is 'modern tool usage with understanding of limitations', where is the limitation?

Ans. Chapter 11's attendance-floor note and chapter 15's Q3: the ladder adapts within a fixed pack; it cannot invent content a specialist educator hasn't validated. That is the stated limitation.

Q. Which single file would you save if the lab machine were wiped tomorrow?

Ans. The four pages plus pack and assets — 30,282 bytes total as logged in chapter 10 — rebuild by running cp08 then cp09; the git-less fallback is this book's listings.

Every figure named in this bank is printed by a capture the previous section quotes; there are no spare numbers in the viva.

Appendix D — Reproducing This Edition, and the Environment

D.1 Reproduction procedure

1. Take the repository as-is (no network needed after Python 3.13 with numpy and pandas exist; weasyprint + PyMuPDF only for rendering this book).
2. Run `python3 collect.py` from the repo root. It generates the field data, executes all fifteen programs plus the generator and the toolchain probe — twice — diffs the two passes byte-wise, refuses on any stderr, and writes out/.
3. Run `python3 make_book_content.py && python3 build.py` to rebuild this PDF; run `python3 qa.py` for the gate. A green QA line ends QA PASSED.

D.2 The collector, as executed for this book

The listing below is the exact file `collect.py` run by the capture; the pre-pass, the pinned environment, the two-pass byte diff and the zero-stderr rule are its whole personality.

```
collect.py

#!/usr/bin/env python3
"""Capture machine evidence for the 1BCP308 Community Project book.

1. src/make_field_data.py runs FIRST as a silent pre-pass (data/ must exist before
any program reads it; it is also captured in its own right, in sort order)
2. every src/<stem>.py : executed with python3 from the repo root with a pinned
environment (PYTHONHASHSEED=0, MPLBACKEND=Agg, LC_ALL=C); command line + stdout
written VERBATIM to out/<stem>.txt; any stderr or nonzero exit fails the capture
3. every src/<stem>.sh : run with sh, captured the same way
4. the WHOLE capture runs twice and the two passes are diffed: any byte difference
refuses the capture (exit 1). Every number quoted in the book is therefore
reproducible on this machine, twice. Programs self-resolve their paths from
__file__, so no argv plumbing is needed here at all.
"""

import os, subprocess, sys
from pathlib import Path

ROOT = Path(__file__).resolve().parent
SRC = ROOT / "src"
OUT = ROOT / "out"
BUILD = ROOT / "build"

ENV = dict(os.environ,
            PYTHONHASHSEED="0", MPLBACKEND="Agg", LC_ALL="C",
            PYTHONIOENCODING="utf-8", PYTHONUNBUFFERED="1",
            MPLCONFIGDIR=str(BUILD / "mpl"))

def run(argv):
    return subprocess.run(argv, cwd=str(ROOT), capture_output=True, text=True, env=ENV)

def one_pass():
    captures, fails = {}, []
    pre = run(["python3", "src/make_field_data.py"]) # silent pre-pass
    if pre.returncode != 0:
        print("FIELD-DATA GENERATION FAILED\n", pre.stderr)
        sys.exit(1)
    for stem in sorted(p.stem for p in SRC.glob("*.py")):
        p = run(["python3", f"src/{stem}.py"])
        lines = [f"$ python3 src/{stem}.py"]
        if p.stderr.strip():
            lines += ["[stderr]", p.stderr.rstrip()]
            fails.append(f"{stem}: stderr not empty")
        if p.returncode != 0:
            lines.append(f"[exit code {p.returncode}]")
            fails.append(f"{stem}: exit {p.returncode}")
        lines.append(p.stdout.rstrip("\n"))
        captures[stem] = "\n".join(lines) + "\n"
```

```

for stem in sorted(p.stem for p in SRC.glob("*.sh")):
    p = run(["sh", str(SRC / f"{stem}.sh")])
    body = (p.stdout + p.stderr).rstrip("\n")
    captures[stem] = f"$ sh src/{stem}.sh\n{body}\n"
    if p.returncode != 0:
        captures[stem] += f"[exit code {p.returncode}]\n"
        fails.append(f"{stem}: exit {p.returncode}")
return captures, fails

def main():
    first, f1 = one_pass()
    second, f2 = one_pass()
    OUT.mkdir(exist_ok=True)
    fails = list(f1)
    for stem, text in first.items():
        if "RESULT" not in text:
            fails.append(f"{stem}: no RESULT marker")
        if second.get(stem) != text:
            fails.append(f"{stem}: NOT REPRODUCIBLE across two runs")
        if text:
            (OUT / f"{stem}.txt").write_text(text)
    n = len(first)
    if fails:
        print("CAPTURE PROBLEMS:")
        for f in fails:
            print(" ", f)
        sys.exit(1)
    summary = (f"RESULT: capture: {n} programs ok, all reproducible across two runs"
              " (zero stderr)")
    print(summary)
    (OUT / "capture_summary.txt").write_text(summary + "\n", encoding="utf-8")

if __name__ == "__main__":
    main()

```

D.3 Field-data generator transcript (the pre-pass, captured)

```

$ python3 src/make_field_data.py
== field-data pre-pass: synthetic, seeded (2025), dependency-embedded ==
wrote data/baseline_survey.csv      rows= 48   2417 bytes
sha256 88812d592080579347096ec038ff4cfff7ec271a07670058c763fbac044dbc4d3
wrote data/pilot_usage.csv          rows=242   3366 bytes
sha256 49a035d67b4cbfb666b5b2591ee5a7c56961ab7bdc27e489fd565977b33afd7a
wrote data/post_survey.csv          rows= 48    998 bytes
sha256 2a51abcc1a34b208980f3ee1c7de3d72774c5afaa00c2c9a968d9a9a4730d444
cohort map: 48 respondents, deterministic learner assignment (i*3 mod 22)
pilot: 22 learners x 12 weeks -> 242 session rows (22 non-attendance cells omitted)
MAKE RESULT: 3 field CSVs written deterministically (seed 2025), sha256 recorded -- PASS

```

D.4 Toolchain and environment (sh probe, captured)

```

$ sh src/toolchain.sh
== CP toolchain / environment capture ==

$ python3 --version
Python 3.13.14

$ python3 -c "import sys; print(sys.executable)"
/usr/local/bin/python3

$ python3 -c "import numpy; print(numpy.__version__)"
2.3.5
$ python3 -c "import pandas; print(pandas.__version__)"
2.2.3

$ uname -srm
Linux 6.1.158+ x86_64

```

```
$ grep PRETTY_NAME /etc/os-release
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"

artifact fingerprints (sha256, first 16 hex) – the book cites these:
syllabus_data.py          963fc7821cc8d148
src/lib/learning_engine.py 6769193fcd2e7080
src/tool/sahaya.css       e41014c7a77dd5b5
src/tool/sahaya.js        97d469f6557b8012
data/baseline_survey.csv  88812d5920805793
data/pilot_usage.csv      49a035d67b4cbfb6
data/post_survey.csv      2a51abcc1a34b208
build/site/pack/content.json 84d679ca54914bcd

note: every capture in this book runs with PYTHONHASHSEED=0 MPLBACKEND=Agg LC_ALL=C

RESULT: toolchain captured
```

D.5 Pedagogy ledger, verbatim rows for weeks this tool serves

From the course page: L:T:P:SL:TW = 0:0:0:30:30 — thirty hours of term work carry the project, thirty of self-learning carry the reading; the capture discipline exists because none of those hours include a proofreader.

Week 8: Selection of final solution and design approach — Concept note / design outline

Week 9: Data processing / calculations / sizing / mapping — Technical analysis sheet

Week 10: Prototype / model / audit / DPR / app development — Working model/draft DPR / map/app

Week 11: Testing, refinement, and implementation planning — Test results / revised output

Week 12: Field implementation / demonstration / awareness activity — Photos, attendance, implementation record

Week 13: Documentation of outcomes, feedback, and impact — Draft report

Week 14: Presentation preparation and viva practice — Presentation deck

Week 15: Final presentation, submission, and reflection — Final report, presentation, viva

About the Author

Dr. Lokesh M R is Professor in the Department of Information Science & Engineering at A J Institute of Engineering and Technology, Mangaluru. His course volumes in this series (operating systems, Java, data structures, EDA, and this community project) share one building rule: every number in the book was produced by a program that was run, captured, and re-run to prove it repeats. He supervises community-project batches with the schools of Dakshina Kannada and maintains that a viva is just an audit with a pulse.